Supplementary Material
SplitGuard-AD: Executable Leakage Auditing for Slice-Based Alzheimer's Disease MRI Deep Learning Benchmarks

S1 Consolidated training hyperparameters

Table S1 lists the full per-cohort training configuration referenced in the main text.

Table S1: Consolidated training hyperparameters across the three tiers and both backbones.

Parameter	Kaggle	OASIS-1	ADNI1
Backbones	ResNet-18; DenseNet-121	ResNet-18; DenseNet-121	ResNet-18 primary; DenseNet-121 architecture arm
Initialisation	ImageNet	ImageNet	ImageNet
Input	$3\times224\times224$	$3\times224\times224$	$3\times224\times224$
Normalisation	ImageNet mean/std	ImageNet mean/std	ImageNet mean/std
Augmentation	flip + $\pm7^\circ$	flip + $\pm7^\circ$	flip + $\pm7^\circ$
Epochs	30	20	15 (all arms)
Optimiser	AdamW	AdamW	AdamW
Learning rate	3×10^{-4}	3×10^{-4}	10^{-4}
LR schedule	Cosine anneal	Cosine anneal	Cosine anneal
Weight decay	10^{-2}	10^{-2}	10^{-2}
Batch size	32	32	16
Loss	inverse-freq BCE	inverse-freq BCE	inverse-freq BCE
Seed array	$\{0, 1, 2, 3, 42\}$	$\{0, 1, 2, 3, 42\}$	$\{0, 1, 2, 3, 4\}$
Backend	CUDA H100	CUDA H100	CUDA H100; RTX PRO 4500 (dose-response matrix); RTX 2000 Ada (volumetric arm); RTX 4090 (exact-linkage, structured-corruption and label-permutation arms)
Mixed precision	AMP	off	off

Every arm ran in one code state on one GPU. Mixed precision is the one setting that differs by cohort: the Tier-1 trainer enables automatic mixed precision on CUDA and the OASIS-1 and ADNI trainers do not. Within any one cohort every protocol is trained under identical settings, which is the condition the inflation-gap comparison requires. The volumetric ADNI arm uses a 3D ResNet-18 on 128^3 skull-stripped, z-scored volumes, batch size 4, AdamW at 10^{-4} with weight decay 10^{-2}, 15 epochs. The exact-linkage, structured-corruption and label-permutation arms use the ADNI1 column unchanged; they vary the manifest or the labels, never the training configuration, which is what lets their contrasts be read against the primary arm.

S2 Operating-point sensitivity table (ADNI)

The clinical translation in the main text is anchored at a screening specificity of 0.90.

The cost-of-leakage arithmetic in full The main text quotes one figure from this translation. The identity is deterministic, missed/1000 = $1000 \cdot \text{prev} \cdot (1 - \text{sens})$, applied to the measured sensitivity at a fixed specificity of 0.90 on the converter-inclusive arm. Table S2 reports it at two prevalence anchors rather than one so that the dependence on prevalence is visible rather than hidden in a single number.

Table S2: Cost-of-leakage arithmetic at two prevalence anchors, converter-inclusive ADNI arm, sensitivity read at fixed specificity 0.90. "Leaky apparent" is what the benchmark would predict; "honest actual" is what the component-safe sensitivity delivers at the same prevalence. The difference is the cost of trusting the benchmark. These are prevalence-weighted re-expressions of a measured sensitivity difference, not deployment estimates.

Prevalence anchor	**Leaky apparent**	**Honest actual**	**Additional missed**
	missed AD diagnoses per 1,000 screened		
US population, 75–84, 13.8% [1]	18.5	64.2	45.7
Memory clinic, 58.9% [2]	79.0	274.0	195.0

The two rows differ by a factor of four, which is the point: the count is a function of the population it is applied to, and quoting one anchor without the other would invite exactly the deployment reading the main text declines.

the table in the main manuscript sweeps that choice, reporting sensitivity at four fixed specificities, specificity at three fixed sensitivities, and the Youden-J point, for each of the three protocols and for the leaky-minus-honest difference. The anchor is not load-bearing: the leaky-minus-honest difference is positive at every one of the ten operating points in the sweep, from +0.099 (specificity at the Youden point) to +0.424 (sensitivity at specificity 0.95), so no conclusion in the main text depends on where the threshold is placed. The magnitude does vary by roughly fourfold across the sweep, which is why the main text quotes one anchor explicitly rather than presenting the difference as a single number.

S3 Mapping SplitGuard-AD outputs to standard reporting domains

Table S4 maps SplitGuard-AD's pre-training audit, model audit, and reporting artefacts to the standard reporting domains covered by TRIPOD+AI and PROBAST domain 4 (Analysis). We deliberately list domains rather than guideline item numbers because TRIPOD+AI numbering is sensitive to the model-development vs. model-validation arm of the guideline; the substantive mapping stays valid regardless of which arm is being checked. SplitGuard-AD is not a replacement for the broader guidelines; it operationalises the subset concerning data partitioning, reproducibility, calibration, and discrimination as machine-checkable outputs.

Domains not directly operationalised by SplitGuard-AD (title/abstract, background and objectives, outcome/predictor variable definition, model specification) remain author- and reviewer-side checklist work; SplitGuard-AD complements rather than substitutes for the broader guideline. PROBAST domain 4 (Analysis) risk-of-bias items concerning sample handling, missing data,

Table S3: Operating-point sensitivity table for the ADNI converter-inclusive arm. **Every threshold is selected on the validation partition and applied to test**; no operating point is chosen by scanning the test ROC. The fixed-rate rows are mean ± SD across five seeds and the Youden rows are seed means, with Δ the Protocol A minus Protocol C difference.

Operating point	Random	Subject-only	Component-safe	Δ
Sensitivity at fixed specificity				
Spec = 0.80	0.919 ± 0.028	0.717 ± 0.122	0.649 ± 0.084	+0.270
Spec = 0.85	0.901 ± 0.024	0.640 ± 0.109	0.616 ± 0.100	+0.285
Spec = 0.90	0.866 ± 0.046	0.551 ± 0.105	0.535 ± 0.103	+0.331
Spec = 0.95	0.810 ± 0.057	0.382 ± 0.055	0.386 ± 0.099	+0.424
Specificity at fixed sensitivity				
Sens = 0.85	0.948 ± 0.028	0.567 ± 0.114	0.563 ± 0.108	+0.385
Sens = 0.90	0.898 ± 0.061	0.439 ± 0.124	0.488 ± 0.141	+0.410
Sens = 0.95	0.782 ± 0.125	0.299 ± 0.111	0.403 ± 0.162	+0.379
Youden-optimal threshold (selected on validation)				
J	0.785	0.458	0.533	+0.252
Sens at J	0.836	0.768	0.682	+0.153
Spec at J	0.950	0.690	0.851	+0.099

and overoptimism in performance estimation are addressed by the same pre-training audit outputs above.

The same outputs answer the CLAIM [3, 4] items a leakage audit touches, listed here by topic rather than by number because the 2024 revision renumbered them. *Data sources and provenance*: the per-image manifest records where each image came from and which identifiers were available, and for Tier 1 the provenance is recovered by content matching rather than assumed (§2.3.1 of the main text). *Data partitioning*: the split rule is frozen in a versioned manifest, the audit reports participant, session and component overlap for each partition, and the compositional check reports how the partitions differ where they are disjoint. *Reference standard*: the label join is audited, with the share of labels resting on a date-proximity match reported alongside the exact-key share. *Evaluation*: performance is reported per protocol with both image-level and participant-level aggregation, calibration, and sensitivity at a fixed specificity. *Failure analysis and reproducibility*: the identity probe, the provenance-degradation sweep and the released rebuild pipeline cover the failure-mode and reproducibility items.

S4 Dose-response mixed-effects model specification

The dose-response OLS slope in the main text (Results: leakage as a dose) is a descriptive calibration constant. To obtain an inferential quantity that respects the paired structure of the 25 (seed, overlap) observations, we additionally fit the random-intercept model

$$\mathrm{AUROC}_{ij} = \beta_0 + \beta_1 p_{ij} + u_i + \varepsilon_{ij}, \quad u_i \sim N(0, \sigma_u^2),\ \varepsilon_{ij} \sim N(0, \sigma^2),$$

by iterated feasible GLS under compound symmetry (`analyze_dose_response_mixed_effects.py` in the release repository). The fitted variance ratio is $\widehat{\rho} = \widehat{\sigma}_u^2/\widehat{\sigma}^2 = 0.317$ on ResNet-18, confirming non-negligible between-seed variation.

Table S4: SplitGuard-AD outputs mapped to standard reporting domains.

Reporting domain	**SplitGuard-AD output**	**Main-text section**
Source of data & provenance	Per-image manifest with subject/session/series IDs and preprocessing lineage; leakage-graph edges over subject/source-volume/near-duplicate.	Methods: cohorts
Eligibility & splitting rules	Component-safe split rule with class-stratification target; orphan-component exclusions audited.	Methods: audit instrument
Sample-size handling & inference	Five-seed array with "descriptive of seed-level variability" labelling and hierarchical (subject $\times$ seed) bootstrap as the inferential cross-check.	Results: statistical conventions
Missing-data audit	Pre-training audit flags unparsable filename keys (Tier 1), missing CDR (Tier 2), and unresolved diagnosis (Tier 3).	Methods: cohorts
Statistical methods	Paired-seed bootstrap with direction-preservation count; hierarchical bootstrap as the defensible inferential quantity.	Results: statistical conventions
Calibration	Brier score and expected calibration error reported across protocols; monotone degradation with leakage protection.	Results: ADNI robustness
Discrimination	AUROC (image- and subject-level), sensitivity at fixed specificity, ROC curves with operating-point overlay.	Results: ADNI arm and screening operating point
Subgroup performance	Sex-stratified AUROC with paired-seed bootstrap (exploratory).	Supplementary S6, Figure S1
Clinical-utility translation	Sensitivity at fixed specificity = 0.90; predictive values at two prevalence anchors.	Results: screening operating point
Reproducibility & versioning	Frozen split manifests (seed-byte-reproducible CSV), versioned audit report, GitHub release.	Data and Code Availability
Limitations & bias declaration	Single-cohort headline scope, $n = 5$ seeds, 2D-slice representation, component-marginal CI straddles zero, subgroup small-sample.	Discussion: Limitations
Code & artefact availability	Apache-2.0 release of manifest builders, leakage-graph construction, split generation, audit, probe, and bootstrap scripts.	Data and Code Availability

Because the design is *balanced* (every seed contributes exactly one observation at each of the five overlap levels) the GLS and OLS point estimates coincide by construction: $\widehat{\beta}_1 = 0.1251$ under both. The random-intercept structure therefore does not move the slope here; what it changes is the uncertainty attached to it, and that is where the inferential content lies.

Inference uses a cluster-robust sandwich standard error over seed with the finite-cluster scaling $\frac{G}{G-1} \cdot \frac{N-1}{N-k}$ and a $t(G-1)$ critical value. With $G = 5$ seeds the correct multiplier is $t_{0.975}(4) = 2.776$, not the normal 1.960; using the normal quantile at this cluster count understates the interval by roughly 40%. The resulting estimates are: ResNet-18 $\widehat{\beta}_1 = 0.1251$, cluster-robust SE 0.0139, 95% CI $[0.0866, 0.1636]$, with a non-parametric seed-cluster bootstrap (B=10,000) giving the comparable interval $[0.1042, 0.1515]$; DenseNet-121 $\widehat{\beta}_1 = 0.0906$, SE 0.0177, 95% CI $[0.0416, 0.1396]$. Both intervals exclude zero, and they overlap substantially, so the two architectures' dose-response slopes are not distinguishable at five seeds.

Spearman ρ on the pooled 25-point (p, AUROC) pairs is 0.941 for ResNet-18 and 0.851 for

DenseNet-121; per-seed AUROC is monotonically non-decreasing in overlap in 5 of 5 ResNet-18 seeds and 0 of 5 DenseNet-121 seeds, consistent with the smaller ResNet-vs-DenseNet inflation-gap asymmetry reported in the ADNI arm.

S5 Metadata linkage-integrity artefact

The full scan-to-visit linkage accounting for the ADNI headline cohort is released as a per-scan CSV (`adni_linkage_audit.csv`, one row per scan), produced by `scripts/build_linkage_audit.py`. Columns:

- `scan_uid`: image identifier (not participant-derived);
- `ptid_hash`: PBKDF2-HMAC-SHA256 of the PTID under a secret per-release salt (600,000 iterations). The salt is never written to the released artefacts;
- `acq_date`: acquisition date, retained because the join cannot be audited without it;
- `join_tier`: `exact_viscode` (direct `(RID, VISCODE)` match), `date_proximity` (visit key failed; label taken from the nearest clinical visit within the 180-day tolerance), or `unresolved`;
- `label_source`: the ADNI column the diagnosis came from;
- `diagnosis_group`: resolved CN / MCI / AD / unknown;
- `in_cnad_universe`: whether the scan enters the CN-vs-AD task.

Tier counts on the released cohort: `exact_viscode` 1,695 (77.7%), `date_proximity` 485 (22.2%), `unresolved` 2 (0.1%). The companion summary (`reports/audits/adni/adni_linkage_audit_summary.json`) records the same counts plus the KDF parameters.

We do not record a candidate-visit count or a minimum date-difference per scan: the manifest builder retains only the tier it resolved to, so those quantities are not recoverable without re-running the join with extra instrumentation. We state this rather than describing fields the released file does not contain.

S6 Sex-stratified performance on ADNI (exploratory)

Stratifying test-set AUROC by sex within each protocol (Figure S1) surfaces a female-minus-male AUROC delta of +0.102 [+0.068, + 0.136] under subject-only and +0.115 [+0.046, +0.181] under component-safe SplitGuard-AD that collapses to −0.020 [−0.055, +0.015] under the leaky protocol, where the sign holds in only 8,893/10,000 resamples. We report the two deltas descriptively and do not test their difference. The gap does not appear to be explained by test-set class imbalance: the female AD share is lower than the male share under all three protocols. Sex differences in AD atrophy trajectories and cognitive reserve are documented in the wider literature [5, 6, 7], but per-cell sample sizes here are small ($\sim$ 8–13 subjects per sex $\times$ age cell) and no adjustment for APOE, education, site, or scanner is performed; **this finding is exploratory and hypothesis-generating only**, and it should not be interpreted as evidence of a biological sex effect without external-cohort replication. The relevant point is methodological rather than biological: leaky evaluation would prevent detection of any such effect if one existed.

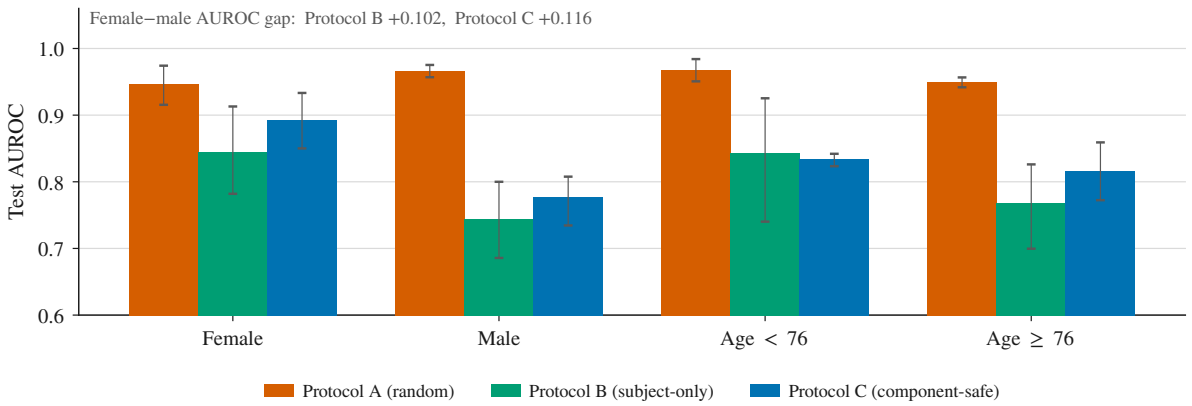


Figure S1: ADNI per-subgroup test AUROC across five seeds (paired-seed bootstrap intervals, $B = 10{,}000$; descriptive). Both honest protocols show a $\sim +0.11$ female-male AUROC gap; under the leaky protocol the same contrast has a point estimate near zero. Reported as exploratory, pending external-cohort replication.

S7 Related work: what each prior treatment of leakage measures

Table S5 places this work against prior treatments of leakage in AD MRI along the six axes the main text uses: whether inflation is shown, whether a dose is measured, whether provenance is varied, whether the mechanism is demonstrated rather than argued, whether damage is observed rather than simulated, and whether split manifests are released.

Table S5: What prior leakage studies establish, and what this paper adds. "Dose" is a measured relationship between the amount of contamination and the reported score, rather than a comparison of two endpoints. "Provenance" is whether the identifiers a protocol needs are treated as a variable rather than assumed. "Mechanism" is whether the inflation is shown to come from patient recall by an intervention that removes the disease signal, rather than argued from saliency or from representation content.

	Inflation shown	Dose measured	Provenance varied	Mechanism shown	Observed damage	Split manifests
Wen *et al.* [8] (review)	✓	–	–	–	–	–
Yagis *et al.* [9]	✓	–	–	–	–	–
Rumala [10]	✓	–	–	saliency	–	–
Rosenblatt *et al.* [11]	✓	three levels	–	–	–	–
Ghiasi *et al.* [12]	✓	–	–	–	–	✓
DataSAIL [13] (tool)	–	–	–	–	–	✓
This paper	✓	continuous	✓	✓	✓	✓

S8 Tier-1 provenance reconstruction: the mapping audit

The main text states that the Tier-1 benchmark is a redistribution of OASIS-1. Because that is a concrete claim about an existing public dataset, it is held to a forensic standard here rather than asserted.

Each Tier-1 image is matched by normalised cross-correlation against every axial slice of every OASIS-1 `t88_masked_gfc` volume. A match is accepted only when one source slice correlates far above every other candidate, so the evidence is the *separation* between the best and second-best match rather than the best match alone. Table S6 reports the audit.

Table S6: Tier-1 to OASIS-1 mapping audit. The decisive row is the separation: the weakest correct match still correlates at 0.9996, while the strongest match to any *other* participant reaches only 0.9623. No threshold between those two values changes any assignment, which is why the mapping is reported as recovered rather than inferred.

Quantity	**Value**
Redistributed images considered	6,400
OASIS-1 volumes searched	436
Images matched	6,400
of which unambiguous	6,400
ambiguous	0
unmatched	0
Source participants recovered	200
Images per recovered participant	32
Axial slice range used	71–102
Participants split across class folders	0
Minimum self-correlation (weakest correct match)	0.9996
Maximum correlation to a different participant	0.9623
Images whose folder label agrees with the source CDR	6,400
Participants whose label agrees with CDR	200 of 200
Recovered participants also present in Tier 2	200 of 200

Two further checks support the reading. The folder labels reproduce the source CDR exactly for all 200 participants, which an accidental or partial correspondence would not do. And every recovered participant is present in Tier 2, so Tier 1 is not an independent cohort and is not counted as one anywhere in the main text.

`scripts/audit_tier1_ground_truth.py` regenerates this table from a reader's own Kaggle and OASIS-1 downloads. The participant-level mapping itself is not redistributed; the script that recovers it is.

S9 Which leakage-graph edge rules fire, per cohort

Table S7 counts, per cohort, how often each edge rule fires and whether it is load-bearing, meaning it merges a pair no other rule had already merged. On the two cohorts with intact identifiers every non-subject rule is redundant; §S4 is unaffected, but the provenance-degradation experiment in the main text exists precisely because that redundancy disappears when identifiers decay.

S10 Audit gate: blocking, reported and specified-but-unenforced checks

Table S8 separates what the released gate enforces from what it reports and from what the framework specifies but does not yet enforce. The distinction matters because an audit whose advertised

Table S7: Which edge rules actually fire, per dataset, counted from the released graph-audit artefacts. The final column is the one that matters for interpreting the component layer: an edge is *load-bearing* only if it merges a pair that no other rule had already merged. On ADNI the session, series-UID and acquisition-date rules fire on 223, 209 and 222 pairs respectively, yet every resulting component still contains exactly one subject, so each of those edges joined a pair that `same_subject` had already joined. They are redundant on these cohorts rather than inert, and become load-bearing only where subject identity is absent, corrupted, or inferred wrongly. The main text's provenance-degradation experiment tests that claim directly by deleting subject identifiers at controlled rates.

Dataset	Edge rule	Pairs joined	Load-bearing here?
ADNI1	`same_subject`	1,800	Yes (sole component-forming rule)
ADNI1	`same_session`	223	No (subsumed by subject)
ADNI1	`same_series_uid` (source volume)	209	No (subsumed by subject)
ADNI1	`same_acq_date`	222	No (subsumed by subject)
267/382 ADNI components are formed by `same_subject` alone; in the other 115 the redundant rules also fire.			
Kaggle	`same_subject` (filename-inferred)	6,076	Fires, but the key is not participant identity
Kaggle	`exact_sha256_duplicate`	0	No pairs found
Kaggle	near-duplicate (dHash ≤ 4)	1,582	Not merged (REVIEW policy)
OASIS-1	`same_subject`	all 416 components	Yes
OASIS-1	`same_session`	all 416 components	No (subsumed by subject)
OASIS-1	`same_source_volume`	all 416 components	No (subsumed by subject)

criteria exceed its implemented criteria is itself an evaluation-integrity failure.

S11 Site and scanner association with label, per partition

Table S9 reports the per-partition association between each audited covariate and the CN/AD label. The audit surfaces the confound; SplitGuard-AD does not attempt to remove it, and the main text states why.

S12 Test-partition composition under the three protocols

Table S10 gives the number of distinct participants, images and images per participant in each protocol's test partition. The three hold image count roughly constant while differing fourfold in how many participants those images come from, which is the asymmetry every protocol-to-protocol comparison in the main text is conditioned on.

References

[1] Kumar B. Rajan, Jennifer Weuve, Lisa L. Barnes, Elizabeth A. McAninch, Robert S. Wilson, and Denis A. Evans. Population estimate of people with clinical Alzheimer's disease and mild cognitive impairment in the United States (2020–2060). *Alzheimer's & Dementia*, 17(12):1966–1975, 2021.

[2] Gavin Thomas, Eric E. Smith, Zahinoor Ismail, Julia G Kirkham, Pamela Roach, Zahra Goodarzi, et al. Potential eligibility for disease modifying therapies among individuals referred

Table S8: GO/NO-GO decision criteria of the released SplitGuard-AD audit (`scripts/audit_adni_splits.py`). We distinguish what the released gate *enforces* from what it *reports* and from what the framework specifies but does not yet enforce, because a leakage audit whose advertised criteria exceed its implemented criteria is itself an evaluation-integrity failure. Only the three blocking checks determine the binary decision; every other quantity is written to the audit report for author and reviewer judgement without halting the pipeline. The gate certifies partition disjointness of the split manifest, *not* site-generalisability, scanner-shift robustness, calibration, fairness, or clinical deployment readiness.

Check	**Status in v1**	**Behaviour**
Subject overlap across partitions	**Blocking**	Any non-blank subject ID in > 1 partition sets `passed=False`; blank and `unknown` IDs are skipped, so a cohort without identifiers cannot pass by vacuity.
Session overlap across partitions	**Blocking**	Any session ID in > 1 partition sets `passed=False`.
Component overlap across partitions	**Blocking**	Any component ID in > 1 partition sets `passed=False`.
Per-partition label balance	Reported	Class counts and ratios written to the report.
Scanner field-strength balance	Reported	Marginal distribution per partition.
Age and sex summaries	Reported	Per-partition demographic summary.
Site–label association (Cramér's V)	Reported	Computed by a separate script (reported with the ADNI results in the main text); not consulted by the gate.
Component-size balance across partitions	Reported	Per-partition component-size distribution, with a warning when partitions are not drawn from comparable participants. Added after auditing this project's own frozen splits, on which it fires (see the main text's Limitations); not consulted by the gate.
Unresolved subject or diagnosis identifiers	**Not enforced in v1**	No check exists. Two ADNI scans with unresolved visit-level diagnosis passed the gate for this reason (see the main text's ADNI cohort description).
Frozen-manifest hash verification	**Not enforced in v1**	Hashed manifests are produced for release but the gate does not re-verify them.
Residual near-duplicate cross-split	**Not enforced in v1**	Near-duplicate edges are applied when *building* components, not re-checked after partitioning.
Within-component label agreement	**Not enforced in v1**	Mixed-label converter components are handled by an explicit pre-specified exclusion rule (stated in the main text's ADNI cohort description) rather than by the gate.

to a tertiary care memory clinic in Calgary, Alberta, Canada: Implications for AD biomarker testing (prospective registry for persons with memory symptoms, PROMPT). *Alzheimer's & Dementia*, 2024.

Table S9: Site and sex confound audit on the five ADNI primary-arm component-safe splits (1,125 manifest rows, 1,123 diagnosis-resolved; CN/AD rows only). Bias-corrected Cramér's V (Bergsma [14]) between each covariate and the binary CN/AD label per partition; mean and range across the five seeds. The site–label association is substantial in every partition and grows as the partition shrinks; it is *exposed* by the audit but not *resolved* by the component-safe split, which is the central caveat of the main text's Limitations. The sex–label association is negligible in train and test but reaches $V = 0.34$ in the small validation partition, an instability we attribute to its size ($n = 165$) rather than to a population effect. Scanner field strength is constant (1.5 T) across every partition of this cohort, so $V(\text{scanner}|\text{label}) = 0$ by construction and the covariate is not tabulated.

Partition	Rows (mean)	Unique sites (range)	V(site|label) (mean [min–max])	V(sex|label)
Train	793	38–39	0.360 [0.347–0.375]	0.000
Val	165	8–10	0.479 [0.464–0.495]	0.343
Test	165	11–12	0.538 [0.418–0.715]	0.046

Scanner field strength is constant (1.5 T) in every partition; $V(\text{scanner}|\text{label}) = 0$ throughout.

Table S10: Test-partition composition per protocol on the ADNI converter-inclusive arm (five-seed means), the arm on which the subject-level aggregation is computed. The three protocols hold image count roughly constant but differ fourfold in how many distinct participants those images come from, because tightening the split rule packs more scans of fewer people into the test partition. Every protocol-to-protocol AUROC comparison in this paper is conditioned on this asymmetry, and we report it here rather than leaving it implicit.

Protocol	**Test participants**	**Test images**	**Images per participant**
A (random)	160.8	213.0	1.32
B (subject-only)	48.0	220.0	4.58
C (component-safe)	38.0	203.2	5.35

[3] Ali S. Tejani, Michail E. Klontzas, Anthony A. Gatti, John T. Mongan, Linda Moy, Seong Ho Park, et al. Checklist for artificial intelligence in medical imaging (CLAIM): 2024 update. *Radiology: Artificial Intelligence*, 6(4):e240300, 2024.

[4] J. Mongan, L. Moy, and C. E. Kahn, Jr. Checklist for artificial intelligence in medical imaging (CLAIM): A guide for authors and reviewers. *Radiology: Artificial Intelligence*, 2(2):e200029, 2020.

[5] E. E. Sundermann, P. M. Maki, L. H. Rubin, R. B. Lipton, S. Landau, and A. Biegon. Female advantage in verbal memory: Evidence of sex-specific cognitive reserve. *Neurology*, 87(18):1916–1924, 2016.

[6] M. M. Mielke, P. Vemuri, and W. A. Rocca. Clinical epidemiology of Alzheimer's disease: assessing sex and gender differences. *Clinical Epidemiology*, 6:37–48, 2014.

[7] X. Hua, D. P. Hibar, S. Lee, A. W. Toga, C. R. Jack, Jr., M. W. Weiner, and P. M. Thompson. Sex and age differences in atrophic rates: an ADNI study with $n = 1368$ MRI scans. *Neurobiology of Aging*, 31(8):1463–1480, 2010.

[8] Junhao Wen, Elina Thibeau-Sutre, Mauricio Diaz-Melo, Jorge Samper-González, Alexandre Routier, Simona Bottani, Didier Dormont, Stanley Durrleman, Ninon Burgos, and Olivier

Colliot. Convolutional neural networks for classification of Alzheimer's disease: Overview and reproducible evaluation. *Medical Image Analysis*, 63:101694, 2020.

[9] Ekin Yagis, Alba García Seco de Herrera, and Luca Citi. Effect of data leakage in brain MRI classification using 2d convolutional neural networks. *Scientific Reports*, 11(1):22544, 2021.

[10] Dewinda J. Rumala. How you split matters: Data leakage and subject characteristics studies in longitudinal brain MRI analysis. In *Clinical Image-Based Procedures, Fairness of AI in Medical Imaging, and Ethical and Philosophical Issues in Medical Imaging (FAIMI 2023)*, volume 14242 of *Lecture Notes in Computer Science*, pages 235–245. Springer Nature Switzerland, 2023.

[11] Matthew Rosenblatt, Link Tejavibulya, Rongtao Jiang, Stephanie Noble, and Dustin Scheinost. Data leakage inflates prediction performance in connectome-based machine learning models. *Nature Communications*, 15(1):1829, 2024.

[12] Mohammad Mahdi Ghiasi, Ryan Stanley Falck, Teresa Liu-Ambrose, and Roger C. Tam. Explainable machine learning for predicting longitudinal dementia status: Establishing a leakage-free benchmark. *PLOS Digital Health*, 5(5):e0001409, 2026.

[13] Roman Joeres, David B Blumenthal, and Olga V Kalinina. Data splitting to avoid information leakage with DataSAIL. *Nature Communications*, 16(3337), 2025.

[14] Wicher Bergsma. A bias-correction for Cramér's V and Tschuprow's T. *Journal of the Korean Statistical Society*, 42(3):323–328, 2013.

[15] S. Kapoor and A. Narayanan. Leakage and the reproducibility crisis in machine-learning-based science. *Patterns*, 4(9):100804, 2023.

[16] M. Roberts, D. Driggs, M. Thorpe, J. Gilbey, M. Yeung, S. Ursprung, A. I. Aviles-Rivero, C. Etmann, C. McCague, L. Beer, J. R. Weir-McCall, Z. Teng, E. Gkrania-Klotsas, J. H. F. Rudd, E. Sala, and C.-B. Schönlieb. Common pitfalls and recommendations for using machine learning to detect and prognosticate for COVID-19 using chest radiographs and CT scans. *Nature Machine Intelligence*, 3:199–217, 2021.

[17] R. Geirhos, J.-H. Jacobsen, C. Michaelis, R. Zemel, W. Brendel, M. Bethge, and F. A. Wichmann. Shortcut learning in deep neural networks. *Nature Machine Intelligence*, 2:665–673, 2020.

[18] L. Oakden-Rayner, J. Dunnmon, G. Carneiro, and C. Ré. Hidden stratification causes clinically meaningful failures in machine learning for medical imaging. In *Proceedings of the ACM Conference on Health, Inference, and Learning (CHIL)*, pages 151–159, 2020.

[19] B. Glocker, R. Robinson, D. C. Castro, Q. Dou, and E. Konukoglu. Machine learning with multi-site imaging data: An empirical study on the impact of scanner effects. Medical Imaging meets NeurIPS Workshop; arXiv:1910.04597, 2019.

[20] W. E. Johnson, C. Li, and A. Rabinovic. Adjusting batch effects in microarray expression data using empirical Bayes methods. *Biostatistics*, 8(1):118–127, 2007.

[21] J.-P. Fortin, D. Parker, B. Tunç, T. Watanabe, M. A. Elliott, K. Ruparel, D. R. Roalf, T. D. Satterthwaite, R. C. Gur, R. E. Gur, R. T. Schultz, R. Verma, and R. T. Shinohara. Harmonization of multi-site diffusion tensor imaging data. *NeuroImage*, 161:149–170, 2017.

[22] Raymond Pomponio, Guray Erus, Mohamad Habes, Jimit Doshi, Dhivya Srinivasan, Elizabeth Mamourian, et al. Harmonization of large MRI datasets for the analysis of brain imaging patterns throughout the lifespan. *NeuroImage*, 208:116450, 2020.

[23] Alzheimer's dataset (4 class of images). Kaggle public dataset, 2019. 6,400 2D brain-MRI JPEG images in four severity folders, uniform 176×208 pixels, released without subject identifiers, acquisition metadata or a statement of origin. Shown in this work by content matching to be 32 axial slices each of 200 OASIS-1 participants, with folder labels equal to their CDR.

[24] Daniel S Marcus, Tracy H Wang, Jamie Parker, John G Csernansky, John C Morris, and Randy L Buckner. Open access series of imaging studies (OASIS): Cross-sectional MRI data in young, middle aged, nondemented, and demented older adults. *Journal of Cognitive Neuroscience*, 19(9):1498–1507, 2007.

[25] P. J. LaMontagne, T. L. S. Benzinger, J. C. Morris, S. Keefe, R. Hornbeck, C. Xiong, E. Grant, J. Hassenstab, K. Moulder, A. G. Vlassenko, M. E. Raichle, C. Cruchaga, and D. Marcus. OASIS-3: Longitudinal neuroimaging, clinical, and cognitive dataset for normal aging and Alzheimer disease. *medRxiv*, 2019.

[26] Michael W. Weiner, Dallas P. Veitch, Paul S. Aisen, Laurel A. Beckett, Nigel J. Cairns, Robert C. Green, et al. The Alzheimer's Disease Neuroimaging Initiative: A review of papers published since its inception. *Alzheimer's & Dementia*, 8(1 Suppl):S1–S68, 2012.

[27] Clifford R. Jack, Matt A. Bernstein, Nick C. Fox, Paul Thompson, Gene Alexander, Danielle Harvey, et al. The Alzheimer's disease neuroimaging initiative (ADNI): MRI methods. *Journal of Magnetic Resonance Imaging*, 27(4):685–691, 2008.

[28] R. C. Petersen, P. S. Aisen, L. A. Beckett, M. C. Donohue, A. C. Gamst, D. J. Harvey, C. R. Jack, Jr., W. J. Jagust, L. M. Shaw, A. W. Toga, J. Q. Trojanowski, and M. W. Weiner. Alzheimer's disease neuroimaging initiative (ADNI): Clinical characterization. *Neurology*, 74(3):201–209, 2010.

[29] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 770–778, 2016.

[30] A. Niculescu-Mizil and R. Caruana. Predicting good probabilities with supervised learning. In *Proceedings of the 22nd International Conference on Machine Learning (ICML)*, pages 625–632, 2005.

[31] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning (ICML)*, volume 70 of *Proceedings of Machine Learning Research*, pages 1321–1330, 2017.

[32] B. Van Calster, D. J. McLernon, M. van Smeden, L. Wynants, and E. W. Steyerberg. Calibration: the Achilles heel of predictive analytics. *BMC Medicine*, 17(1):230, 2019.

[33] Vishnu M. Bashyam, Jimit Doshi, Guray Erus, Dhivya Srinivasan, Ahmed Abdulkadir, Ashish Singh, et al. Deep generative medical image harmonization for improving cross-site generalization in deep learning predictors. *Journal of Magnetic Resonance Imaging*, 55(3):908–916, 2022.

[34] Lemuel Puglisi, Arman Eshaghi, Geoffrey J M Parker, Frederik Barkhof, Daniel C Alexander, and Daniele Ravi. DeepBrainPrint: A novel contrastive framework for brain MRI re-identification. *Proceedings of Machine Learning Research, Medical Imaging with Deep Learning*, 2023.

[35] C. Wachinger, P. Golland, W. Kremen, B. Fischl, and M. Reuter. BrainPrint: A discriminative characterization of brain morphology. *NeuroImage*, 109:232–248, 2015.

[36] Pouria Rouzrokh, Bardia Khosravi, Shahriar Faghani, Mana Moassefi, Diana Vera Vázquez, Yashbir Singh, and Bradley J. Erickson. Mitigating bias in radiology machine learning: 1. data handling. *Radiology: Artificial Intelligence*, 4(5):e210290, 2022.

[37] Clifford R. Jack, David A. Bennett, Kaj Blennow, Maria C. Carrillo, Billy Dunn, Samantha Budd Haeberlein, et al. NIA-AA research framework: Toward a biological definition of Alzheimer's disease. *Alzheimer's & Dementia*, 14(4):535–562, 2018.

[38] J. S. Lin, E. O'Connor, R. C. Rossom, L. A. Perdue, and E. Eckstrom. Screening for cognitive impairment in older adults: A systematic review for the U.S. Preventive Services Task Force. *Annals of Internal Medicine*, 159(9):601–612, 2013.

[39] R. Wittenberg, M. Knapp, B. Hu, A. Comas-Herrera, D. King, A. Rehill, C. Shi, S. Banerjee, A. Patel, C. Jagger, and A. Kingston. The costs of dementia in England. *International Journal of Geriatric Psychiatry*, 34(7):1095–1103, 2019.

[40] Elias Chaibub Neto, Abhishek Pratap, Thanneer M. Perumal, Meghasyam Tummalacherla, Brian M. Bot, Andrew D. Trister, Stephen H. Friend, Lara Mangravite, and Larsson Omberg. Learning disease vs participant signatures: a permutation test approach to detect identity confounding in machine learning diagnostic applications, 2017.

S13 Extended methods and results

The main text reports the measurement and the evidence that carries it. This appendix holds the full treatment of each component: the cohort construction, the audit instrument, the arms and probes that support the headline, and the analyses the main text summarises in a sentence. Nothing here is new; it is the same material at the length it was written, retained so that every reported quantity remains traceable.

S14 A leakage taxonomy for slice-based AD MRI

We enumerate nine leakage and shortcut modes relevant to AD MRI deep learning. The list is intended as an operational checklist for AD MRI benchmarks rather than a closed taxonomy of every conceivable shortcut; the modes below are drawn from the prior leakage and shortcut literature [8, 9, 15, 16, 17, 18] and consolidated here into a structured form whose risk labels are calibrated to AD MRI deep-learning practice. Table S11 provides a structured overview. The new element relative to the existing leakage taxonomies is the explicit treatment of the near-duplicate (image-content) mode and the longitudinal-converter mode as first-class entries: both are characteristic of AD MRI workflows (augmented public releases, multi-visit research cohorts) and are not separately enumerated in general-purpose taxonomies such as Kapoor and Narayanan [15].

Subject, session, longitudinal, and slice leakage all violate the same clinical independence assumption: a test case should represent a new patient, not another view, visit, or derivative of a patient already seen during training. This is especially important for 2D MRI pipelines because repeated slices from one 3D scan preserve head shape, intensity profile, slice position, and preprocessing traces that can function as biometric identifiers. Near-duplicate leakage is related but broader: visually similar derived images may cross splits even when filenames or metadata no longer expose their shared origin.

Source, site/scanner, and demographic shortcuts are not leakage in the narrow train-test contamination sense, but they can produce equally misleading clinical estimates. If label prevalence differs by dataset source, scanner protocol, age, sex, or disease severity, a model can learn cohort structure instead of disease-relevant neuroanatomy. Finally, clinical-label shortcut occurs when a benchmark mostly separates advanced atrophy from healthy controls; such performance may be real for that narrow severity contrast while still failing to support early diagnosis or deployment claims.

Figure S2 visualises the four leakage modes that share the train–test-contamination structure and that motivate the edge design of the SplitGuard-AD leakage graph in §S15. The empirical decomposition in §S20.0.1 shows that the dominant mode on the AD MRI cohorts studied here is longitudinal same-subject leakage (panel b), and that subject-only splitting alone (Protocol B) captures $\sim 93\%$ of the correction on the ADNI headline cohort. On Tier 1 that protocol is not available: grouping by the only identifier the release carries still leaves 70% of the test images with a participant seen in training (§S16.0.1). The additional component-safe layer (Protocol C) contributes a smaller marginal above that baseline, because on cohorts with intact subject identifiers the two protocols produce the same partition. What separates them is what happens when those identifiers are not intact: §S23 measures the divergence directly, and finds that Protocol C prevents 49% of the residual patient leakage Protocol B admits once a tenth of the identifiers are gone.

S15 The audit instrument

SplitGuard-AD is an open-source reference implementation and audit pipeline, not a general-purpose library: it comprises five sequential modules (Figure S3).

S15.0.1 Metadata registry

Each image is assigned a row in a manifest with fields including `image_id`, `subject_id`, `subject_id_confidence`, `raw_class_label`, `binary_label`, `source_dataset`, and preprocessing lineage. For public JPEG benchmarks lacking original DICOM metadata, subject IDs are inferred from filename conventions with confidence flags for unparseable cases; Tier 1 shows that such inference can be wrong in ways no confidence flag detects (§S16.0.1).

S15.0.2 Leakage-graph construction

A leakage graph $G = (V, E)$ is constructed over the image set V. We distinguish the edge families that are *blocking* (those whose endpoints are merged into one component and therefore can never be split) from those that are *reported* for inspection, because the released implementation treats them differently and the distinction changes what the framework guarantees.

Blocking edges. An edge (u, v) is added, and the endpoints merged, if u and v share an inferred subject identifier, a session identifier, a source volume (identified by series UID), an acquisition date

within a subject, or are byte-identical (equal SHA-256). Which of these are available is cohort-dependent, and the released audit reports the per-reason edge counts so a reader can see which rules actually fired: the ADNI tier supplies subject, session, series-UID, longitudinal and acquisition-date identifiers; the public-benchmark tier has no acquisition provenance at all, so only inferred subject and exact-SHA edges exist there. Supplementary Table S5 gives the per-cohort breakdown.

The practical consequence is stated plainly: on the two source cohorts the connected components coincide with participant grouping, and on Tier 1 they coincide with an identifier that is not participant identity. The additional edge families are implemented, auditable and do fire, but none of them changes a single component here. Establishing their value requires a cohort whose leakage structure departs from subject identity (§S26).

Reported edges. Near-duplicate pairs are detected with a difference hash (dHash) at 16×16, i.e. a 256-bit code, and flagged when the Hamming distance satisfies $d \leq \delta$ with $\delta = 4$. Under the default policy (`--near-duplicate-policy review`) these pairs are written to a candidate file and surfaced in the audit report but are *not* merged; this is the policy under which every manifest and every number in this paper was produced. A `blocking` policy that additionally merges them is implemented and switchable. We keep review as the default because the identities recovered for Tier 1 (§S16.0.1) let this threshold be calibrated rather than assumed, and the calibration shows that no setting of it is safe to gate on. At $d \leq 4$ the detector is precise against recovered identity (precision 0.986) but sees only 1.6% of same-participant pairs, so merging its firings moves the Tier-1 test contamination from 70.0% to 61.2%: real, and nowhere near a fix. At $d \leq 8$, where it recovers 82% of adjacent-slice pairs, its 8,636 firings merge the 6,400 images into 27 components, so the split stops separating anything at all. A further 1,343 of those firings join images carrying opposite labels and are refused rather than merged, because the splitter requires label-pure components: the rule is simultaneously too aggressive to split on and unable to act on a sixth of what it finds. A rule with that shape belongs in an audit report a human acts on, not in an automatic gate.

We report the hash family and width explicitly because they are not interchangeable: dHash at 256 bits with $\delta = 4$ is a far stricter criterion than the 64-bit perceptual hashes for which the $\delta = 4$ convention was originally proposed. Connected components $C_1, \ldots, C_k$ of G define the atomic units for splitting: no component may be split across train, validation, and test partitions.

S15.0.3 Component-safe split generation

Given target split ratios $r_{\text{train}} : r_{\text{val}} : r_{\text{test}}$ and a class stratification target, SplitGuard-AD assigns entire components to splits by stratified assignment against image-count targets, seeded for reproducibility: within each stratum it fills the validation target with whole components, then the test target from what remains, and leaves the rest to training. Targets are image counts rather than component counts, and the test target is the remainder after train and validation are rounded, so the three never over-subscribe the stratum. On Tier 1 the subset choice is an exact dynamic-programming solve; on ADNI it is a greedy descending-size fill approaching the target from below. Ties are broken on a frozen seeded rank, which is what makes the manifests both byte-reproducible and genuinely seed-sensitive. The resulting split manifest is frozen as a CSV file and versioned; no training script may discover or modify splits at runtime. the algorithm in the main manuscript formalises the full procedure: leakage graph construction (lines 1–8) followed by component-safe assignment (lines 9–18). All randomness is funnelled through the single SEED argument, so the manifest is byte-reproducible.

Implementation note the algorithm in the main manuscript shows the double loop in $\mathcal{O}(N^2)$ form for clarity; exact identifier edges are implemented by hash-grouping in $\mathcal{O}(N)$. The near-

duplicate comparison is the exception, evaluated over all $\binom{N}{2}$ pairs, which is tractable at the scale studied here ($N \leq 6{,}400$) and is the component that would need locality-sensitive indexing at 10^5 images or more.

Protocol nomenclature Throughout the experiments we evaluate three split protocols on the same image universe:

Protocol A (Random, leaky). Random image-level shuffle into train/val/test; the conventional benchmark setup.

Protocol B (Subject-only). Split by the supplied subject identity: all images of one subject land in one partition, with no component or near-duplicate constraint. Reported where it isolates a distinct layer, chiefly the ADNI experiments of §S19.

Protocol C (Component-safe, SplitGuard-AD). Full leakage-graph component-safe split: subject identity, scanner-session re-acquisitions, and near-duplicate components are all respected.

The three protocols form a strict ordering (A $\supseteq$ B $\supseteq$ C) in terms of leakage permissiveness; the three-point degradation analysis in §S20.0.1 traces the contribution of each layer.

S15.0.4 Pre-training audit

Before training, SplitGuard-AD emits a structured audit report with three parts: (i) component overlap across partitions; (ii) label distribution per partition; and (iii) association between label and each audited covariate, given as Cramér's V per partition. The covariates are site code (extracted from the ADNI `PTID`), scanner field strength and sex.

This last part is complementary to scanner-effect work [19] and to harmonisation in the ComBat lineage [20, 21, 22]. SplitGuard-AD surfaces such confounds but does not remove them, because the gap measured here is dominated by the subject-identity channel, which post-hoc harmonisation does not address. The clinical claim-boundary statement in the report card (main manuscript) is authored, not emitted by the audit. The audit report issues a binary GO / NO-GO decision for training according to the formal criteria in Supplementary Table S6.

S15.0.5 Model audit layer

After training, SplitGuard-AD evaluates trained model representations with a linear probe: probe accuracy above chance indicates residual shortcut encoding. In this release only the *subject-identity* probe is implemented and reported (§S20, `scripts/run_biometric_probe_adni.py`), because subject identity is the channel the component-safe split is designed to close. Probes for dataset source, site/scanner, and demographic attributes follow the same construction and are the natural extension of this layer, but are not implemented in v1 and no such probe result is reported here; the site–label association we do report (§S19) is a metadata-level contingency statistic, not a representation-level probe.

S16 Cohorts

Cohort roles Two source cohorts carry the measurements. The **ADNI1: Complete 3Yr 1.5T** collection (§S16.0.3) is the paper's *headline* cohort: all headline numerical claims in the abstract,

the cost-of-leakage analysis, the subgroup analysis, and the dose-response stress test derive from it. **OASIS-1** (§S16.0.2) is the replication cohort, where participant identifiers are authoritative. It is operationally external to ADNI: a different recruiting site, scanners, acquisition protocols and recruitment era. The direction-preservation we report between the two cohorts is therefore a partial cross-site generalisation check, not external validation in the regulatory sense. The third tier is not a third cohort. The **Kaggle public JPEG benchmark** (§S16.0.1) is, as we show by content matching, a redistribution of OASIS-1 itself whose participant identifiers have been replaced by a filename index that both merges and splits participants. We therefore use it for what it documents rather than for what it would replicate: an observed and measurable instance of the provenance damage that §S23 otherwise has to simulate. Throughout the experiments we use the shorthand *Tier 1* (redistributed public benchmark), *Tier 2* (OASIS-1), *Tier 3* (ADNI1 headline) for brevity.

S16.0.1 Tier 1: a redistributed public benchmark

The widely distributed public Kaggle Alzheimer's MRI benchmark dataset [23] comprises 6,400 JPEG images of uniform 176×208 pixels in four severity classes: NonDemented ($n = 3{,}200$), VeryMildDemented ($n = 2{,}240$), MildDemented ($n = 896$), and ModerateDemented ($n = 64$). The release states no origin, carries no acquisition metadata, and exposes subject identity only through a parenthesised filename index (*e.g.*, `NonDemented_patient100_1 (100).jpg`).

What the images are We recovered the origin by content. Every image was matched against axial slices of all 436 OASIS-1 cross-sectional volumes in their processed, atlas-registered, masked form (`T88_111/*_masked_gfc`; `scripts/audit_tier1_ground_truth.py`). All 6,400 images match unambiguously: the lowest correlation between an image and its matched slice is 0.9996, while the highest correlation any image reaches with a slice of a *different* participant is 0.9623. The benchmark is 200 OASIS-1 participants at 32 consecutive axial slices each (atlas slices 71–102). The folder labels reproduce those participants' Clinical Dementia Rating exactly, for every one of the 6,400 images (CDR = 0 → NonDemented, 0.5 → VeryMild, 1 → Mild, 2 → Moderate), and no participant's slices are split across two folders: they are CDR, not the weak folder-derived labels they are usually assumed to be. Every one of the 200 participants is also in our Tier-2 OASIS-1 set, so Tier 1 and Tier 2 are two views of one cohort, and the cross-tier replication in this paper rests on two source cohorts rather than three.

What redistribution destroyed The filename index is not a participant identifier. Its 324 distinct keys divide into 149 that correspond to exactly one participant and 175 that merge two or three participants under a single key, while 179 of the 200 participants are spread across two or more keys and one is spread across 32. A further 128 images (2.0%) carry no parsable index at all. Grouping by this key is therefore not subject-wise splitting: it is grouping on a key that is wrong in both directions at once, and the supplied identifier alone does not reveal that. Recovering it required matching the images against the source cohort, which a user of the release is not placed to do.

Against the recovered identities the damage is exact. In the frozen seed-42 split used throughout this paper, which groups by the filename key, 85 participants have images in more than one partition, 46 of the 58 test participants also appear in training, and 647 of the 960 test images (67.4%) belong to a participant the model was trained on. the table in the main manuscript repeats that measurement for every grouping rule the release supports.

Near-duplicate structure, read against ground truth No exact SHA-256 duplicates exist. The near-duplicate detector (dHash at 16×16, a 256-bit code, Hamming $d \leq 4$) flags 1,582 candidate pairs, of which 1,559 (98.5%) join two slices of the same participant and 1,526 join slices one position apart in the same volume. Of the 57 pairs that straddle the frozen partitions, the recovered identities make 52 the same participant and 5 different. The filename key had called 36 of those 57 "different subjects", and in 31 of them the key was wrong and the detector right. Read against the release's own identifier the detector looks like a 63% false-positive rate; read against ground truth its cross-partition output is 91% true leakage. An audit that can only compare a detector against the identifiers it already holds will systematically mistake a broken identifier for a false alarm.

Calibrating the detector With every pair's identity known the threshold is calibrated over all 20,476,800 pairs rather than argued for. At $d \leq 4$ it is precise and sparse (precision 0.986, recall 0.246 of adjacent-slice pairs, 0.016 of all same-participant pairs, false-positive rate 1.1×10^{-6}); at $d \leq 8$ it is sensitive and unusable as a blocking rule (precision 0.699, recall 0.816, false-positive rate 1.3×10^{-4}, the 8,636 pairs that collapse the graph in the table in the main manuscript). The median Hamming distance between two slices of one participant is 33 bits against 35 between participants, so the detector separates neighbouring slices, not participants, and perceptual hashing cannot stand in for a participant identifier.

S16.0.2 Tier 2: OASIS-1

We use OASIS-1 as the metadata-rich replication cohort [24]; LaMontagne *et al.* [25] describe the successor OASIS-3 longitudinal release for context. All 12 official raw cross-sectional archives were available locally, together with the demographic/clinical and reliability spreadsheets. The clinical spreadsheet contains 436 session rows from 416 unique subjects, including 20 MR2 reliability sessions. For each session, SplitGuard-AD selectively extracts one processed masked T88 MPRAGE Analyze pair (`PROCESSED/MPRAGE/T88_111/*_masked_gfc.img/.hdr`) without fully expanding the raw archives.

CDR is used as the primary clinical label: CDR = 0 is mapped to NonDemented, and CDR > 0 to Demented. Sessions with missing CDR are retained in the manifest for auditability but excluded from the primary classification experiment. The resulting OASIS-1 manifest contains 135 NonDemented sessions, 100 Demented sessions, and 201 sessions with missing CDR. Five central coronal slices are derived from each processed volume, yielding 2,180 total slices and 1,175 labelled slices (675 NonDemented, 500 Demented). The frozen OASIS-1 SplitGuard-AD split contains 825/175/175 labelled slices in train/validation/test, with 165/35/35 subjects respectively and zero subject or component overlap across splits.

S16.0.3 Tier 3: ADNI1 Complete 3Yr 1.5T (headline cohort)

We use the ADNI1: Complete 3Yr 1.5T image collection as the paper's headline cohort, accessed under an active LONI IDA Data Use Agreement; the consortium overview [26], the MRI-acquisition protocol [27], and the clinical-characterisation core [28] provide the foundational methodology references for the ADNI release used here. All headline numerical claims in the abstract, the intro, and the cost-of-leakage and dose-response analyses derive from this cohort. The collection contains 2,182 pre-processed 1.5T T1-weighted volumes (MPR or MPR-R variants with GradWarp, B1 correction, N3, and intensity scaling already applied) from 382 unique subjects spanning the ADNI1/GO/2/3/4 phases (the table in the main manuscript). A small subset (28 scans, 17 of

which fall in the CN/AD universe) are magnetization-transfer T1 (MT1) acquisitions rather than MPRAGE; these are retained in the primary analysis and excluded in a pre-specified sensitivity arm (§S19).

Diagnosis labels are taken from the ADNI diagnostic summary table (`DXSUM_PDXCONV_ADNIALL`) under a phase-aware visit-level mapping. Every visit in the export we use carries the canonical `DIAGNOSIS` code (1=CN, 2=MCI, 3=AD), and all 2,180 resolved labels in the released manifest come from it (`label_source=DIAGNOSIS`). The released builder also implements two fallbacks for exports that differ from ours: a longitudinal `DXCHANGE` collapse to the destination state, and the mutually exclusive ADNI1 legacy flags `DXAD`, `DXMCI`, `DXNORM`, which some ADNI1 case report forms use instead of one categorical field. Neither fallback fires on this export, which has no `DXCHANGE` column at all, so neither contributes a label reported here; we describe them because they are in the released code, not because they shaped these results. Scan-to-visit alignment uses a 180-day date-proximity join through the MRI metadata table, with a documented date-only fallback when the viscode-keyed lookup fails. The full mapping rule and edge-case handling are pre-specified before any model training; the resulting manifest yields 756 CN, 768 MCI, and 656 AD scans, plus two scans whose diagnosis cannot be resolved (kept in the manifest with `diagnosis_group=unknown` for auditability and excluded from the primary CN-vs-AD task).

Linkage-integrity audit Every ADNI label rests on this join, so it is accounted for per scan in `adni_linkage_audit.csv` (`scripts/build_linkage_audit.py`; schema and provenance in Supplementary Material S5). Of the 2,182 scans, 1,695 (77.7%) were labelled through a direct `(RID, VISCODE)` match, 485 (22.2%) through the 180-day date-proximity fallback, and 2 (0.1%) remained unresolved. The primary arm is the worst case: of its 1,123 CN-vs-AD scans, 841 carry the exact visit key and 282 (25.1%) the date fallback, against 297 of 1,412 (21.0%, with 1,115 on the exact key) in the whole CN-vs-AD universe, because the converter components this arm excludes are almost fully covered by the visit key (5.2% date-only). That one label in four rests on a date match is a limitation of the label chain, so the exact-linkage arm of §S19 retrains all three protocols on the 841, keeping the same 220 participants, and asks whether the headline gap survives without the weaker quarter.

For apples-to-apples comparison with Tier 2, each volume is reduced to a single 2D coronal-centre slice. The slice is computed once at ingest with 1st–99th percentile intensity normalisation and saved as a single-channel PNG; this caching step is byte-identical to the volume-time computation in §S16.0.2 but fits the full collection in 54 MB rather than 77 GB of NIfTI volumes. The leakage graph links scans sharing a subject identifier, a session identifier, a series UID, or an acquisition date; these four rules fire on 1,800, 223, 209 and 222 pairs respectively. The ADNI builder computes no image-hash edges, by design and not by omission, so this tier carries no near-duplicate protection: that channel is measured on Tier 1, where the images are public and may be hashed and redistributed. The 382-subject collection collapses into 382 single-subject components (mean 5.7 scans per component, range 4–11), so the frozen SplitGuard-AD split is effectively a subject-stratified partition with additional session-, series- and date-level edges that, on this cohort, every one of which is subsumed by the subject rule (§S23).

The CN-vs-AD binary task universe excludes 768 MCI scans by construction (the binary task is cross-sectional, not longitudinal progression-prediction) and a further 289 scans (216 AD + 73 CN) drawn from components tagged `mixed_label`: subjects whose visit sequence crosses CN, MCI, and AD diagnoses (longitudinal converters). Including those subjects in the primary arm would require either (i) breaking the subject-level partition (the same subject contributes CN visits to one split and AD visits to another), or (ii) collapsing their longitudinal trajectory to a single subject-

level label, which conflates cross-sectional CN-vs-AD classification with progression-prediction. We adopt the cleaner option: exclude these 289 scans from the primary arm and re-include them in a pre-specified sensitivity arm (§S19, "Sensitivity to converter exclusion"), assigning each mixed-label component a CN-or-AD majority label for stratification only. No scan is relabelled: training and evaluation use each scan's own visit-level diagnosis throughout, so a converting participant contributes CN scans as CN and AD scans as AD within one component. The total inflation gap in the sensitivity arm (+0.142) is *larger* than in the primary arm (+0.118), so the exclusion does not drive the headline result. Removing these 289 scans leaves 1,123 CN/AD scans. The frozen split manifests carry 1,125 rows, because two scans whose visit-level diagnosis did not resolve (`diagnosis_group=unknown`, §S16.0.3) are kept in the manifest and flagged there rather than deleted from it. They are not, however, evaluated: every trainer maps only CN and AD to a target and silently skips any other value, so the two rows enter no partition the model sees. The evaluated universe is therefore 1,123 scans across 220 single-subject components, equivalently 220 participants, since every component here contains exactly one subject. Under component-safe splitting each frozen seed produces train/validation/test partitions of 793/165/165 evaluated scans; the manifest row counts are two higher. We record the discrepancy rather than reconciling it silently, because a manifest row count and an evaluated scan count are different quantities and conflating them is how a cohort size drifts. Figure S4 summarises the end-to-end pipeline from the 10 LONI IDA archive zips through the slice-cached manifest to the per-seed splits.

S17 Observed provenance damage on a redistributed benchmark

Setup Tier 1 is trained under three protocols that differ only in what they group by. Protocol A shuffles images at random. Protocol B′ groups by the filename key, the only identifier the release carries and therefore the strongest protocol an audit run from inside the release can implement. Protocol C′ groups by the participant identity recovered in §S16.0.1. Everything else is held fixed. The backbone is ResNet-18 [29] with ImageNet pre-trained weights, trained for 30 epochs at batch size 32 with random horizontal flip and $\pm 7^\circ$ rotation. The optimiser is AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$, AMSGrad disabled) at learning rate 3×10^{-4} and weight decay 10^{-2}, with cosine annealing and a class-frequency-weighted BCE loss. The seed array is $\{42, 0, 1, 2, 3\}$.

Leakage verification Under Protocol A, 195 of 219 test-set filename keys (89.0%) also appear in training. Under Protocol B′ no filename key crosses a partition, so an audit restricted to the release's own metadata reports zero leakage, while 70.0% of the test images in fact belong to a participant the model trained on. Under Protocol C′ the contamination is zero (the table in the main manuscript).

Results Random splitting inflates test AUROC from 0.766 to 0.999, a gap of +0.233. That part is the familiar result. The part this tier exists for is the middle column: grouping by the release's own identifier recovers 88.6% of that gap and leaves +0.027 AUROC of inflation standing, because the key it groups on merges and splits participants. A practitioner who does everything correctly with the information the benchmark provides still reports an inflated number, and nothing inside the benchmark reveals it. This is the difference between leakage as a protocol error, which the literature already documents, and leakage as a provenance failure, which survives a correctly applied protocol. Figure S5 separates what is stable across seeds from what is not. Protocol A sits above both grouped protocols in all five seeds (Figure S5a) and on every metric at seed = 42 (Figure S5b), so the headline gap is not a seed artefact. The residual between B′ and C′ is the

smaller quantity and it is not uniform: their AUROC ordering reverses in one seed of five (the table in the main manuscript), and at seed = 42 C′ is the more specific of the two while B′ is the more sensitive, which is what a threshold applied to a less separable score produces. The +0.027 residual is therefore a five-seed mean and not a per-run guarantee, and we hold it to the same standard as the ADNI component marginal rather than a softer one: its subject × seed interval is $[-0.058, +0.113]$, which spans zero, exactly as ADNI's 0.008 [-0.033, +0.042] does. Both are unresolved at five seeds and we report both that way. What the Tier-1 tier establishes is therefore the *contamination*, which is structural and needs no interval – 70% of test images share a participant with training under the release's own key – not the AUROC consequence of that contamination, which these five seeds cannot pin down. The distinction matters because the contamination is what a practitioner can check and the AUROC consequence is what they would have to take on faith.

What the three test sets are not A random split spreads a participant's 32 slices over all three partitions, so its test partition holds many more participants, each contributing far fewer images, than the participant-grouped one. That asymmetry is a property of the protocols and cannot be removed by resampling: subsampling participants leaves AUROC essentially unchanged, because it is a rank statistic over pairs. What it does mean is that the three AUROCs are estimated over differently composed test sets, and we report the composition rather than adjusting for it (§S19, Supplementary Table S8).

Learning curves Figure S6 shows validation AUROC per epoch. Protocol A saturates within the first third of training, which is the signature of memorising images it will be tested on rather than learning a disease feature; Protocol B′ tracks it at a lower level, still lifted by the participants its key failed to group; Protocol C′ converges lowest and slowest.

Tier-1 takeaway Tier 1 is not an independent cohort and we no longer treat it as one: it is 200 OASIS-1 participants redistributed without their identifiers. That makes it the one place in this paper where provenance damage is observed rather than simulated, and it supplies the quantity the simulation in §S23 is calibrated against: an identifier that is wrong in both directions leaves +0.027 AUROC of inflation in place after a correctly applied subject-wise split. The mechanism behind that inflation is examined in §S20, the layer-by-layer decomposition in §S20.0.1, and the cohorts with intact identifiers in §S18 (OASIS-1) and §S19 (ADNI).

S18 Replication on OASIS-1

We repeat the leaky-versus-SplitGuard-AD comparison on OASIS-1 using the same ResNet-18 family and the derived 2D slice representation described in §S16.0.2. Protocol A randomly splits the 1,175 labelled slices by image; Protocol C uses the frozen subject-component-safe SplitGuard-AD split. Across five random seeds, Protocol A places nearly every test subject in training (mean test-subject contamination $99.54\% \pm 0.68\%$, range 98.5–100%).

Why the OASIS-1 gap is mechanically smaller than ADNI's Before we report the AUROC numbers, we surface the structural properties of the OASIS-1 slice manifest that bound the leakage surface *a priori* (script: `scripts/oasis_subject_level_summary.py`; JSON: `reports/tables/oasis1_subject_level_summary.json`). Across the labelled subset ($n = 235$ subjects with binary NonDemented/Demented labels; a further 181 subjects are dropped for missing CDR at the labelling stage) OASIS-1 has 5 coronal slices per session and 2,180 slices across 436 sessions. Sessions per

subject average 1.05: only 20 subjects, 4.8% of the 416 imaged, carry a repeat session. Per-subject image counts are in fact comparable between the two cohorts (OASIS-1 $\approx$ 5.2 slices per subject; ADNI1 primary arm $\approx$ 5.1 scans per subject at one coronal-centre slice per scan, §S16.0.3), so the smaller OASIS-1 gap is not attributable to a thinner per-subject image stack. The more plausible mechanism is structural: only 4.8% of OASIS-1 subjects carry a repeat session, so the cohort contains almost no longitudinal within-subject variation for a model to exploit, whereas the ADNI1 cohort is longitudinal by construction. The OASIS-1 replication is therefore best read as a directional replication of the mechanism, not as a high-powered replication of the inflation magnitude reported on ADNI (§S19). The OASIS-1 gap is reported at the image level, and should be read as the weaker of the two cross-cohort quantities when quoted alongside the ADNI subject-level result.

The OASIS-1 result (the table in the main manuscript) validates the core leakage mechanism on the cohort Tier 1 was drawn from: the leaky protocol still places nearly every test subject in the training set (99.54%). While the operating-point metrics (F1, sensitivity) are statistically noisy across five seeds due to the smaller dataset size, and one seed produces a negative AUROC gap, the contamination result confirms that random slice splitting creates severe subject leakage even when true metadata are available.

DenseNet-121 architecture replication Under a DenseNet-121 backbone and the same five seeds the OASIS-1 gap is +0.048, mirroring the ResNet-18 result above in point estimate and in seed-to-seed noise on this small cross-sectional cohort.

S19 Replication on ADNI1, and its robustness arms

We repeat the leaky-versus-SplitGuard-AD comparison on the ADNI cohort introduced in §S16.0.3, using the same ResNet-18 family and coronal-centre-slice representation as §S18. Three split protocols are compared on the same 220-participant CN-vs-AD manifest of 1,125 rows, of which 1,123 are diagnosis-resolved. Protocol A splits images at random 70/15/15 and is the leaky protocol. Protocol B partitions by subject 70/15/15, without the leakage-graph component, series or near-duplicate constraints. Protocol C uses the frozen SplitGuard-AD component-safe split. Test partitions hold 169, 169 and 165 scans respectively, because the component-safe subset-sum assignment cannot hit the target size exactly. Each protocol is run for five seeds, 15 epochs each, with AdamW at lr $= 10^{-4}$ and a class-balanced BCE-with-logits objective.

Under Protocol A, the random image-level split places nearly every test subject into the training partition: the mean test-subject contamination across the five seeds is 99.7%, with several seeds reaching 100%. Protocols B and C, by construction, achieve 0% test-subject overlap.

the table in the main manuscript reports the per-protocol test AUROC with paired-seed bootstrap intervals (B=10,000 resamples of the five-seed array, descriptive; see "Statistical conventions" at the start of the main manuscript). The total inflation gap (Protocol A − Protocol C) is +0.118 [+0.096, +0.145] AUROC, and the leaky->-SplitGuard-AD direction holds in all five seeds. A two-level hierarchical (subject $\times$ seed) re-analysis, the more defensible inferential quantity, gives +0.118 with an interval of [+0.061, +0.181] (see Robustness analyses below). The gap decomposes into a large subject-identity component (Protocol A − Protocol B, 0.110) and a small marginal beyond patient-level splitting (Protocol B − Protocol C, 0.008 [-0.033, +0.042]) whose interval touches zero.

Subject-level vs image-level evaluation (recommended clinical reading) An image-level AUROC weights participants by scan count, and participants do not contribute equal numbers of scans. We therefore aggregate each participant's per-scan predictions (mean y_{prob}) and recompute AUROC at the participant level on the converter-inclusive arm (the table in the main manuscript): the gap survives the aggregation, and per-protocol AUROC is marginally higher under it, consistent with within-participant noise reduction. The image-level number is retained as the cross-cohort-comparable headline; *but the subject-level number is the recommended reading for clinical interpretation*, because it is a stricter aggregation that rules out per-subject scan-count heterogeneity as a confound and is closer to the patient-level unit of clinical decision-making. Downstream users of SplitGuard-AD report cards should quote the subject-level gap in clinical-adjacent contexts and the image-level gap only in benchmark-comparison contexts.

Robustness analyses Four pre-specified arms test whether the headline gap depends on a particular methodological choice, a fifth row reports the post-hoc size-balanced control of §S26, and a sixth isolates the label-linkage question raised in §S16.0.3 (the table in the main manuscript). All four preserve the leaky->-SplitGuard-AD ordering in every paired-seed bootstrap resample, and the total gap moves by at most 0.028 AUROC across them. The component marginal does not behave that way. It is Protocol B − Protocol C throughout, and it inverts to -0.026 on the converter-inclusive arm, opposite to the direction a leakage-removal effect would take, since there the component-safe constraints raised AUROC relative to subject-only splitting rather than lowering it. No interval excludes zero in any arm. We therefore report the total gap as robust and the component marginal as unresolved at five seeds, and we do not resolve it by adding seeds. Two results say why. §S23 shows the margin is near-zero on intact cohorts by construction, so the informative axis is provenance loss rather than seed count; and §S28 measures what a spurious contrast of this kind looks like at this seed count, finding one of +0.094 between two protocols that had nothing to learn.

The sixth row answers the label-linkage question directly. Restricting the cohort to the 841 scans whose diagnosis comes from an exact visit key, which discards a quarter of the primary arm's scans while keeping all 220 participants, gives a total gap of +0.117 against the primary arm's +0.118, positive on all five seeds, and a component marginal of +0.024 whose interval again spans zero. The headline gap is therefore not an artefact of the date-matched labels. We put that no more strongly than the arm supports: dropping a quarter of the scans widens the interval by nearly half again, from [+0.096, +0.145] to [+0.072, +0.166], so the two point estimates agreeing to 0.001 is a coincidence inside overlapping intervals rather than a precise replication. What the arm establishes is consistency, and that the date-matched quarter is not carrying the result. The arm also completes a sensitivity analysis that §S26 could only run halfway. That one rescored the primary arm's test partition on its exact-key scans, which tests the labels the gap is measured *against* while the date-matched scans stay in training; here they are absent from both sides, and the answer does not change.

The DenseNet-121 arm additionally replicates on OASIS-1 (+0.048, §S18), so the inflation is consistent in direction across two convolutional backbones on both source cohorts, though on OASIS-1 neither backbone's hierarchical interval excludes zero. A two-level subject × seed hierarchical resampling widens the total-gap interval on every arm, and we give all of them rather than only the primary one: [+0.061, +0.181] (primary), [+0.081, +0.208] (converter-inclusive), [+0.072, +0.191] (MT1 excluded), [+0.087, +0.209] (DenseNet-121), [+0.070, +0.209] (size-balanced), [+0.145, +0.335] (volumetric) and [+0.046, +0.195] (exact label linkage). All seven exclude zero. The two OASIS-1 arms are the exception and are reported with the cross-cohort

comparison in §S18: $[-0.033, +0.107]$ on ResNet-18 and $[-0.021, +0.121]$ on DenseNet-121, both spanning zero. Calibration is far worse under both honest protocols than under the leaky one, and not ordered between them (Brier/ECE: Protocol A 0.089/0.079, Protocol B 0.247/0.253, Protocol C 0.196/0.188) [30, 31]: the leaky protocol is calibrated to its own contaminated test distribution, which is not a property worth having [32]. Test-set demographic composition is balanced by construction across the five frozen splits (mean 50%±6% female, 49%±7% aged ≥ 76; 220 participants, median age 76).

Whole-volume input The volumetric arm asks whether the shortcut belongs to the 2D slice or to the participant. A 3D ResNet-18 on 128^3 volumes, trained on the same 220 participants under the same three protocols, puts the total gap at +0.235 [+0.174, +0.298], positive in 5/5 seeds and $[+0.145, +0.335]$ under the subject × seed hierarchical resampling, against +0.118 on the 2D primary arm. We report the two side by side rather than as a ratio: the recipes differ in pretraining and augmentation as well as in dimensionality, so the comparison establishes that the shortcut survives whole-volume input, not that volumes leak twice as much. The gap widens from both ends, and only one of those ends is about leakage. The leaky level rises to 0.971, against 0.947 in 2D, which is what a memorisable unit of one whole volume rather than one slice predicts. The honest level falls to 0.736, against 0.829 in 2D, and that is not a leakage effect: this network is trained from scratch on 703–723 volumes per run, where the 2D arms fine-tune ImageNet features, so its honest performance is not a competitive volumetric baseline and we do not offer it as one. The transferable quantity is the gap and not the level, here as everywhere else in this paper. The decomposition reproduces the 2D pattern: the subject layer carries the inflation (+0.268 $[+0.163, +0.372]$) while the component layer adds nothing resolvable at five seeds (-0.033 $[-0.185, +0.122]$, spanning zero, with subject-only at 0.703 falling below component-safe just as it does in the converter-inclusive 2D arm).

Site × label confound check The ADNI CN/AD universe spans 48 recruiting sites (site code extracted from the first triplet of the ADNI `PTID`); on the full universe, uncorrected Cramér's V between site and CN/AD label is 0.36 ($\chi^2 = 178$, $p < 0.001$), reflecting the well-known fact that ADNI sites differ systematically in their CN/AD subject mix. Within the five frozen component-safe SplitGuard-AD splits, this association is preserved rather than eliminated (script: `scripts/site_scanner_confound_audit.py`; JSON: `reports/tables/adni/adni_site_scanner_confound_audit.json`). Supplementary Table S7 reports bias-corrected Cramér's V (Bergsma's small-sample correction [14]) for site×label and sex×label across the five primary-arm component-safe splits. The site–label association is moderate in train ($V \approx 0.36$), substantial in validation ($V \approx 0.48$), and strong in test ($V \approx 0.54$, ranging to 0.72 across seeds); the sex–label association is negligible in train and test but reaches $V \approx 0.34$ in the small validation partition. Scanner field strength is constant across this cohort and so cannot confound the label. The framework does not perform site-aware partitioning by design and is therefore not a substitute for site harmonisation (ComBat [20, 21, 22, 33]) or for explicit site-stratified splitting; the inflation gap we report is the subject-identity channel that harmonisation does not fix post-hoc (§S15). A site-stratified extension of the component-safe construction is named as future work in §S26.

Figure S7 places the ADNI gap in context under one reporting convention, paired-seed bootstrap intervals at $B = 10{,}000$. The gap is positive in the mean on both source cohorts: the small cross-sectional one (OASIS-1, +0.035 [+0.009, +0.064]) and the longitudinal one with substantial repeat structure (ADNI, +0.118 [+0.096, +0.145]). It is positive again, at +0.233 [+0.191, +0.270] (hierarchical $[+0.159, +0.311]$), on the redistribution of the first, where each participant

contributes 32 slices rather than five. The three differ in how firmly, and we state the difference rather than averaging over it. On ADNI and Tier 1 the sign holds in every seed and both the paired-seed and the hierarchical intervals exclude zero. On OASIS-1 it does not: one of five seeds is negative (-0.010), and while the paired-seed interval excludes zero the subject $\times$ seed hierarchical interval is $[-0.033, +0.107]$, with the sign preserved in 84% of resamples. OASIS-1 is therefore a directional replication at the level of the seed mean, not an independently significant one, and we read it that way throughout. Patient identity is the dominant leakage channel on the two cohorts where a subject-only protocol exists to decompose it, ADNI and Tier 1; OASIS-1 carries no such arm. Subject to that, the random-split benchmark overstates AD-classification performance by a magnitude robust to the seed array, the converter-inclusion rule, the acquisition type, the CNN backbone, input dimensionality and the strictness of the label linkage.

the figure in the main manuscript consolidates the headline inflation results across both source cohorts, the redistributed benchmark, both backbones and every ADNI robustness arm in a single reporting frame.

S20 Is patient identity decodable, and does that explain the gap?

The permutation control shows the shortcut is taken. A natural next question is what makes it available, and the answer is a structural property of the representation rather than anything about the protocol. A linear probe (LinearSVC) trained on frozen penultimate-layer features (512-dim) of the 15 ADNI checkpoints of the converter-inclusive arm recovers participant identity far above chance, and to the same degree whichever protocol trained the representation: 211.7$\times$ chance under component-safe splitting. Holding out a whole visit per participant instead of a random fifth of their images, so that no repeat acquisition from the same session sits on both sides, leaves identity decodable at 204.8$\times$ chance. We run the probe on ADNI rather than on Tier 1 because ADNI supplies the participant identifiers the probe classifies; on Tier 1 the release's key is not participant identity (§S16.0.1).

The interpretation is narrower than the number invites. Identity is decodable from every one of these representations, including those trained under honest protocols, so the probe cannot explain a *difference* between protocols: a quantity that is the same in both conditions cannot account for the AUROC gap between them. What it establishes is a structural property of 2D MRI. Repeated slices of one head share shape, intensity profile and slice position, so identity is available to any visual representation, and that availability is what makes test-set contamination exploitable. This also qualifies the reading that saliency evidence invites: Rumala [10] observes that a leaky network attends to subject-identifying structure and takes that as the mechanism of the inflation, but identity in the representation is a precondition for the shortcut rather than evidence that it was taken. The permutation control of §S28 supplied that evidence by a route needing no saliency map, and the probe explains why the route exists: identity is there to be used. The probe uses one architecture and one probe family; a non-linear adversary in the manner of DeepBrainPrint [34] or BrainPrint [35] would upper-bound it. The causal weight therefore rests on the three controlled interventions of §S28, §S22 and §S23.

S20.0.1 Three-point degradation curve

The probe answers *whether* identity is encoded; the degradation curve answers *how much* of the inflation each grouping removes (Figure S8).

The A$\rightarrow$B$'$ step is ΔAUROC $= +0.206$ and the B$'$ $\rightarrow$C$'$ step a further $+0.027$. On ADNI, where identifiers are intact, the same decomposition puts $\sim 93\%$ of the gap in the first step. The

two cohorts bracket the claim: grouping on a correct identifier removes most of the inflation, and grouping on a damaged one does not.

Decomposition takeaway (and head-to-head against GroupKFold) The three-point degradation isolates $\sim 93\%$ of the ADNI inflation gap to subject-identity transfer; on Tier 1 the release's own identifier recovers 88.6% of the gap, the remainder being the participants that identifier fails to group (§S17). Protocol B enforces the same subject-independence constraint as `GroupKFold`, under a fixed three-way split rather than K-fold cross-validation (main manuscript); we did not run `GroupKFold` itself, so this decomposition quantifies the marginal value of the component-safe layer above subject-grouped partitioning rather than constituting a head-to-head tool benchmark. That marginal, reported throughout as Protocol B − Protocol C, is 0.008 [-0.033, +0.042] on the ADNI primary arm and -0.026 on the converter-inclusive arm (the table in the main manuscript): small, of inconsistent sign, and with no interval excluding zero. On Tier 1 the comparison does not exist in that form, because the release carries no identifier that Protocol B could group on.

That marginal should be read with a further caveat that we did not appreciate until auditing our own splits. Protocols B and C do not differ only in their grouping. Protocol B shuffles participants and cuts at 70/15/15 by participant count; Protocol C consumes a frozen manifest produced by a size-descending greedy bin-packer that hits 70/15/15 by image count. The two therefore differ in the grouping (the variable of interest) and also in the assignment algorithm and the quantity the ratios are computed over. On a difference that small, the assignment step is not obviously negligible relative to the grouping step. The same control failure, measured directly in §S23, was worth about half the apparent effect there. We therefore report the component marginal as an upper bound on what the grouping contributes rather than as an isolated estimate of it, which reinforces rather than weakens the conclusion already stated: the marginal is small, of inconsistent sign, and not resolved.

The greedy also leaves a visible fingerprint on the frozen ADNI manifests. Because it fills the validation and test bins first and takes the largest components that still fit, the partitions end up nearly disjoint in component size: across the five seeds, training components span 4–6 scans (mean 4.65), validation 7–9 (mean 7.50) and test 5–7 (mean 6.11). Participants with longer longitudinal records are systematically routed away from training. Class balance survives this, because components are stratified by label before the subset-sum assignment runs (39.1%, 38.2% and 40.0% AD by image across the three partitions), but within each class the test partition is enriched for participants with more follow-up. That matters clinically: in this cohort AD participants contribute fewer scans than CN participants (4.55 ± 0.97 against 5.56 ± 1.00), so scan count tracks retention, and retention tracks disease course. This is an instance of the duration-dependent selection bias described in the data-handling literature [36], arriving through the splitter rather than through the cohort.

We do not leave this as a caveat. The frozen manifests are retained because every number in this paper is computed on them and exact reproducibility is worth more than a cosmetic correction, but a size-balanced split was generated specifically to test whether the headline result depends on the defect. Visiting components in shuffled rather than size-descending order removes the stratification. Mean component sizes become 5.15, 5.00 and 5.07 scans in train, validation and test, against 4.65, 7.50 and 6.11 on the frozen split. The total gap is +0.136 on the size-balanced split against +0.118 on the frozen one, with hierarchical intervals $[+0.070, +0.209]$ and [+0.061, +0.181]. We report it beside the primary arm rather than after it. The composition defect is real, it was found by this project's own audit of its own splits, and the effect it was suspected of producing is not there. What it does mean for the audit gate, which passed this split on every check it performs, is taken

up in §S26.

The practical recommendation on a cohort like this one is therefore the familiar one, split by participant identity. The component and near-duplicate layers cost almost nothing here and buy nothing measurable either, and we claim no AUROC benefit from them on ADNI or OASIS-1. What they buy appears on the axis where it can, once identifiers begin to disappear (§S23). Readers running the framework on a cohort with clean, converter-free identities should expect the component-safe and subject-only protocols to be indistinguishable, and should read that as the instrument reporting correctly rather than as a disappointment.

Where grouping by the available identifier is blind Subject-wise splitting can only protect images whose participant is known, and a redistributed benchmark is exactly the case where that premise fails without saying so. Tier 1 makes the failure measurable, because its true identities are recoverable from outside the release (§S16.0.1), which is not something its users have.

The exposure is not a handful of edge cases but the structure of the release, measured in §S16.0.1: a key that merges and splits participants, images with no key at all, and a split that every check the release supports calls clean while most of its test set is contaminated. This is the regime the component layer exists for, and the honest reading of Tier 1 is that the layer does not rescue it either, because the redistribution removed the session and series identifiers the graph would have carried. The consequence for a reader of public benchmarks is stronger than a recommendation to split by subject: where a release does not document provenance, a subject-wise split of it cannot be verified from inside the release at all, and the number it reports cannot be interpreted.

S21 What the gap costs at a screening operating point

Running both protocols on the same universe surfaces two findings the leaky benchmark conceals. A sex-related performance disparity emerges under both honest protocols and collapses to indistinguishable from zero under the leaky one (Supplementary Material S6); sex differences in AD are well documented [7, 5, 6], and we report ours as exploratory without attributing it to any of those mechanisms. The gap also translates into a screening-operating-point cost that a reader of the leaky number would underestimate (§S21.0.1). Both rest on the per-image predictions persisted by the converter-inclusion arm.

S21.0.1 Cost-of-leakage as an arithmetic illustration

To express the gap in units a clinician can weigh, we re-expressed the sensitivity difference at a fixed screening specificity of 0.90 as missed-diagnosis counts per 1,000 screened, under literature-cited prevalence anchors. Sensitivity at that specificity is 0.866 ± 0.046 under Protocol A against 0.535 ± 0.103 under Protocol C, which at the US 75–84 population prevalence of 13.8% [1] is 45.7 additional missed AD cases per 1,000 screened if the leaky benchmark is trusted.

This is deterministic arithmetic applied to the measured sensitivities, not a deployment estimate. It inherits every cohort caveat in §S26: a single 2D coronal-centre slice, a research cohort that is not a screening population, a threshold chosen on validation, and no prospective evaluation. We give it because an AUROC difference is hard to weigh and a count is not, and we give it once. The operating-point derivation, the sensitivity of the result to the chosen specificity, and the second prevalence anchor are in Supplementary Material S2.

S22 Leakage as a dose: injecting test-set overlap

The cross-tier inflation gap and the cost-of-leakage translation both take the leakage condition as binary (leaky vs. honest). A more informative question for benchmark designers is the dose-response: if p% of test subjects also appear in train, how much AUROC inflation should we expect? We answer this with a controlled stress-test on the converter-inclusive ADNI sensitivity arm.

Design Starting from the Protocol C component-safe split, we generate a family of injected splits at target contamination fractions $p \in \{0, 0.25, 0.50, 0.75, 1.0\}$, where p *is the share of test scans that belong to a training participant*, not the share of test participants. For each non-zero p we draw $k = \lfloor p \cdot 258 \rceil$ participants from the multi-visit training subset and substitute one alternate scan of each into the test set, displacing an equal number of original test scans. Because each donor contributes a single scan, the share of contaminated test *participants* is never smaller and is usually far larger (63%, 77% and 86% of test participants at $p = 0.25$, 0.50 and 0.75, converging to equality only at $p = 1$), which is why every composition below is done in scan units. Class balance of the *injected partition* (118 CN / 140 AD over 258 rows) is held fixed at every overlap level by stratified substitution. The partition the model is *evaluated* on is not held fixed either, and this qualifies the calibration. The binary pipeline drops MCI scans and substitution displaces them, so the evaluated CN/AD subset grows from 203 rows at $p = 0$ to 258 at $p = 1$, with the AD share rising from 47.7% to 54.3%. These are five-seed means; the $p = 0$ count runs from 197 to 207 across seeds.

The training partition is not held fixed either, and this is the more consequential of the two. Each substituted scan is moved out of training, so evaluated training rows fall from 1,041 at $p = 0$ to 783 at $p = 1$, 24.8% of them, as the dose rises. The donor *participant* remains in training, which is why donors are drawn only from the multi-visit subset, so the identity leakage the arm is built to create is genuine; what the arm does not do is hold the training set constant while adding it.

The dose axis is therefore confounded with both test composition and training size, and the two plausibly act in opposite directions: thinning the training set should depress AUROC at high p, while the class enrichment above has no clear sign. The slope is consequently descriptive of this substitution procedure as a whole, and is not a causal price per unit of realised leakage. Both quantities are measured by `scripts/analyze_dose_response_composition.py` and reported in `reports/tables/adni/adni_dose_response_composition.json`. We report it rather than correct it because the two cannot be separated without changing the injection mechanism, and because the two cannot be separated without changing the mechanism: the slope is used only as a floor (§S21.0.1). Otherwise this is biometric-identity leakage alone: no slice or near-duplicate is shared between train and test, only the patient. We run the full 5 seeds × 5 overlaps matrix on two architecture families (ResNet-18 and DenseNet-121, 15 epochs each, lr = 10^{-4} AdamW), a total of 50 training runs.

Result Figure S9(a) shows the resulting per-seed AUROCs and the seed-averaged dose-response curves with 95% paired-bootstrap CIs. The seed-averaged curves increase with test-subject overlap for both architectures; per-seed monotonicity holds in 5/5 ResNet-18 seeds but in 0/5 DenseNet-121 seeds (Supplementary S4); the fitted ResNet-18 endpoints (0.823 at $p = 0$, 0.948 at $p = 1$) sit within 0.02 of the converter-inclusive arm's honest and leaky levels (0.815 and 0.957, §S19), which is the arm the injected splits are built from. We read that agreement as a sanity check and not as a calibration: the matrix and those levels come from the same code state on different GPUs (§Declarations), and nothing in the injection was tuned to reproduce them. The straight-line OLS

fit (Figure S9(b)) recovers

$$\widehat{\text{AUROC}}_{\text{ResNet-18}} = 0.823 + 0.125 \cdot p, \qquad R^2 = 0.853, \tag{S1}$$

and a similar but noisier curve for DenseNet-121 ($\widehat{\text{AUROC}} = 0.861 + 0.091 \cdot p,\ R^2 = 0.64$). The slope of +0.125 AUROC per unit overlap on ResNet-18 implies a cohort- and architecture-specific calibration point: on this ADNI universe and this 2D-slice representation, every 10 percentage points of test-subject overlap inflates the reported AUROC by approximately +0.013. The point-estimate predictions from Equation S1 are usefully read as the deterministic part of a noisy calibration; the per-seed scatter around the regression line in Figure S9(b) is approximately consistent with an additive Gaussian residual, so the dose-response can equivalently be written

$$\text{AUROC}(p) \ = \ \alpha + \beta \cdot p \ + \ \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \sigma^2), \tag{S2}$$

with $(\widehat{\alpha}, \widehat{\beta}) = (0.823, 0.125)$ and $\widehat{\sigma} = 0.0191$ on ResNet-18 (0.0250 on DenseNet-121), the residual standard deviation of the twenty-five per-seed observations about the OLS fit. The unit matters: because the band is meant to describe a *single* training run, it has to be estimated from single runs. Taking it instead from the five per-overlap means would give 0.008 and understate the spread roughly 2.5-fold, since averaging five seeds removes most of the seed-level variance the band exists to represent. So estimated, $\widehat{\sigma}$ gives an order-of-magnitude prediction band for one run at a given p and lets the equation be used probabilistically (*e.g.*, the probability that a benchmark reporting AUROC A at measured contamination p exceeds its true component-safe value by more than 0.05 can be estimated from the upper tail of $\mathcal{N}(A - \widehat{\alpha} - \widehat{\beta}p,\ \widehat{\sigma}^2)$). We report the OLS slope as a descriptive calibration constant rather than as an inferential coefficient (the twenty-five (seed, overlap) observations are not independent under OLS). A random-intercept model fitted by iterated feasible GLS under compound symmetry gives the same slope, $\widehat{\beta} = 0.1251$, because the design is balanced; what changes is the interval. With a cluster-robust sandwich standard error carrying the finite-cluster correction and a $t(4)$ critical value, the 95% CI is $[0.0866, 0.1636]$ on ResNet-18; DenseNet-121 gives $\widehat{\beta} = 0.0906$ with 95% CI $[0.0416, 0.1396]$. Both exclude zero. Full specification, the fitted variance ratio, the seed-cluster bootstrap, and Spearman monotonicity statistics are in Supplementary Material S4. Density-choice qualifications and finer-grid trade-offs are discussed in §S26.

Interpretation The response is approximately linear across the whole $[0, 1]$ range, with a mild saturation near the leaky end: the ResNet-18 increments shrink across the sweep (+0.046, +0.035, +0.025, +0.020) and the DenseNet-121 mean dips at the last step (0.9403 at $p = 0.75$ to 0.9387 at $p = 1$). That is what a bounded metric does near its ceiling, not a threshold below which contamination is harmless; no threshold appears anywhere in the range.

Both architectures converge at $p = 1$ on the Protocol A level, which is the same convergence the deletion sweep produces from the other direction: total subject overlap and absent identifiers are two routes to the same condition. The DenseNet-121 curve starts higher (0.853 against 0.814 at $p = 0$) and is flatter (+0.091 against +0.125), but the mixed-effects slope intervals overlap ($[0.0866, 0.1636]$ against $[0.0416, 0.1396]$; Supplementary S4), its R^2 of 0.64 trails the ResNet-18 fit and no DenseNet seed is monotonic across all five overlaps. We run no difference test and draw no inference about backbone capacity: at five seeds the trend is carried by the seed average rather than visible within a seed.

Practical use, and why the slope is a floor Equation S1 converts a measured test-subject contamination fraction (which the pre-training audit already reports) into expected AUROC inflation. It should be read as a lower bound on the gap rather than as an estimate of it, for a reason

that follows from how it was fitted: the injection varies test-subject overlap by substitution, so it prices the participant-identity channel and no other, while thinning the training set as it does so. A naive split on a real cohort additionally leaks through slices of one volume, near-duplicates and re-acquisitions, none of which the slope sees.

Two of the three cohorts bear this out and the third does not; we give all three. Under random splitting contamination is essentially complete on each, including 99.5% of OASIS-1 test subjects, so the equation predicts a drop of at most 0.125. Measured gaps are +0.142 on the converter-inclusive ADNI1 arm the injected splits are built from, +0.233 on Tier 1, where adjacent slices of one volume add a channel the slope does not price, and +0.118 on the primary arm, which the floor matches to within 0.01. OASIS-1 does not fit: its contamination is as complete yet its gap is +0.035, roughly a third of the prediction. We read that as the boundary of the calibration rather than a refutation, since OASIS-1 contributes five slices from one cross-sectional visit while the slope was fitted where repeat visits dominate. Used as a point estimate the equation would overstate a reader's honest AUROC, which is the wrong direction for a quantity meant to correct over-reporting; used as a floor it answers whether the contamination present is large enough to matter.

S23 Provenance loss: what the leakage graph buys when identifiers fail

§S15.0.2 reports that on every cohort studied here the connected components coincide exactly with a subject grouping, so the component layer earns nothing over `GroupKFold`-style splitting. That is a consequence of subject being the coarsest identifier: session, series and acquisition-date edges all nest inside it, so none of them can change a component while `subject_id` is present and correct. The framework's stated scope condition is therefore that the extra edge families matter only where subject identity is absent, corrupted, or inferred wrongly. Until now that was an assertion. This section measures it.

Design The regime is created deliberately, using the same template as the dose-response experiment in §S22: hold everything fixed, sweep one controlled quantity, read off the curve. Here the swept quantity is provenance loss. Starting from the ADNI1 primary arm, where subject identifiers are complete and authoritative, we delete `subject_id` on a random 0/10/25/50/100% of scans and ask what each protocol does with the remainder. Deletion is applied per scan rather than per subject because that is the observed failure mode: redistributed benchmarks lose the patient index one filename at a time, so a record is typically partly recoverable rather than wholly anonymous. Under Protocol B every scan whose identifier was removed becomes its own group and may be placed in any partition. Under Protocol C the leakage graph is rebuilt from whatever identifiers survive; `session_id` and `series_uid` are globally unique on this cohort (1,003 and 1,008 distinct values, no blanks, and no value shared between two participants) and do not take `subject_id` as an input, so they continue to bind scans that `same_subject` can no longer reach.

The outcome is scored against the untouched ground-truth identifiers, which the degradation preserves out of band. We report the number of true participants with scans on both sides of a partition boundary. This is a structural measurement: it counts the leakage each protocol actually admits without routing the question through a trained model, and it is therefore free of the seed-variance and test-set-size caveats that qualify the AUROC comparisons elsewhere in this paper.

Result the figure in the main manuscript and the table in the main manuscript give the curve. At zero loss the two protocols admit no leakage and are indistinguishable, reproducing the equivalence

reported in §S15.0.2. They separate immediately thereafter: at 10% identifier loss the leakage graph prevents 49% of the participant leakage subject-wise splitting admits (9.4 versus 18.6 straddling participants), and 30–37% at higher loss rates.

Both protocols are assigned by the same class-stratified bin-packer over their respective groupings, so the difference is attributable to the grouping and not to the assignment step. That control matters more than it may appear. An earlier version of this comparison assigned Protocol B by shuffling subject groups and cutting at 70/15/15 by group count, while Protocol C bin-packed by image count; under that mismatched design the graph appeared to prevent 78% of residual leakage at 10% loss. Roughly half of the apparent benefit came from the assigner rather than from the grouping. We report the matched comparison because only the grouping is the claim.

At the end of the sweep the protocol becomes its own failure mode The structural counts above are what the split admits. What the experiment reports is an AUROC, and following it to the end of the sweep gives the result this paper is built on.

With identifiers complete, subject-wise splitting on this arm reports 0.771 $[0.709, 0.833]$. With every identifier removed, the same procedure, unchanged, reports 0.946 $[0.921, 0.972]$. The random image-level split of §S19 reports 0.947 $[0.926, 0.965]$ on the same cohort. The difference between the last two is 0.001, and their intervals overlap across almost their whole length.

We state that as a failure to distinguish rather than as an equality: five seeds cannot establish that two procedures agree, and overlapping intervals are weak evidence. What the comparison does establish is that nothing in the reported number separates them. A reader handed the right-hand column cannot tell from it that the split was meant to be patient-disjoint, and neither can the procedure, which raises no error: every scan is a singleton, every singleton is its own group, and grouping by group is vacuous. The inflation across the sweep is +0.176 AUROC, and it accrues without a warning at any point.

That is the sense in which patient-wise splitting is not a property of a splitting algorithm. The algorithm is unchanged along this axis. Only the evidence it is given changes, and the guarantee it is credited with degrades continuously into the condition it was adopted to prevent.

Provenance loss amplifies non-linearly Deleting 10% of identifiers returns 36% of test participants to the training partition under Protocol B, a roughly threefold amplification. The mechanism is elementary once stated: a participant contributes a median of five scans, and one of them losing its index is enough to detach that scan from the group and let it land opposite the other four. Participant-level protection is not degraded gracefully by scan-level provenance loss; the first missing identifier in a record breaks it.

Composing the two curves, and where the composition fails Chaining the two curves requires them to share a unit. The injection experiment sweeps the share of contaminated test *scans*, so that is the quantity we compose through: at 10% identifier loss it is 35% under Protocol B and 18% under Protocol C, against 36% of test *participants* under Protocol B. Read through Equation S1, that is +0.043 AUROC of expected optimism under Protocol B against +0.022 under Protocol C. That composition under-predicts what the same experiment measures directly, and the comparison is worth stating because it bounds how much weight the chain carries: the provenance arm itself trains at every deletion level, and its test AUROC rises from 0.771 at no loss to 0.865 under subject-wise splitting and 0.829 under the component-safe protocol at 10% loss, that is +0.095 and +0.059 against the two composed figures just given. The chain is low by rather more than a factor of two in both protocols. Two reasons are visible: the slope is fitted on a cohort whose

contamination is varied by substitution, which thins training as it adds overlap, whereas deleting identifiers perturbs both sides differently, and the slope and the shares come from different arms of one code state (§Declarations). We therefore read the composition as an order of magnitude and a lower bound, never as a coefficient, and we report the directly measured rise beside it rather than only the chain. the table in the main manuscript reports the participant share p alongside it, because that is what a split audit shows a reader.

We do not present that as a general result, because p is not monotone across the sweep. At 25% loss Protocol C's p (0.408) exceeds Protocol B's (0.374), inverting an ordering that holds at every other level and that the straddling-participant count preserves throughout. The reason is that p is a ratio whose denominator is the number of distinct test participants, and the two protocols do not produce test partitions containing the same number of them: component-safe grouping yields larger and fewer components, hence a smaller and differently composed test partition. p is therefore a defensible input to the dose-response fit, where it is measured within a single protocol, but a poor instrument for comparing protocols. The straddling-participant count, which counts events rather than ratios, does not have that defect, and it is the measurement the result above rests on. We give the composition at 10% as an illustration of what chaining the two experiments would yield, and explicitly not as a validated protocol-level comparison.

Scope Three limits belong with this result. The measurement is primarily structural: it counts admitted leakage, and the AUROC arm trained on the same degraded manifests is reported in §S26 rather than here. The degradation is synthetic, imposed on a cohort with complete provenance rather than observed on one that lost it, which is what makes ground-truth scoring possible and also what makes it a model of the failure rather than the failure itself. And the recovery depends on which identifiers a cohort happens to carry: ADNI1 supplies globally unique session and series keys, whereas the Tier-1 benchmark supplies neither, so the curve measured here is an upper bound on what the same protocol could achieve on a cohort stripped down to filenames.

S24 Structured provenance corruption: where the graph earns its place

The deletion experiment removes identifiers *visibly*. A missing `subject_id` is a value a splitter can see is absent, and the honest response, treating that scan as its own group, is available to any `GroupKFold` user. Redistribution rarely fails that politely. The Tier-1 benchmark in §S17 carries a key for every image and the key is simply wrong: it splits 179 of 200 participants across several keys and merges others under one. Nothing is missing, so nothing looks broken.

Design We therefore apply a corruption operator $C \in \{\textsc{drop}, \textsc{split}, \textsc{merge}\}$ at intensity λ to the ADNI1 primary manifest, rebuild both groupings on the corrupted result, and score residual leakage against untouched ground truth. DROP removes a scan's identifier with probability λ; SPLIT deals a participant's scans into two pseudo-identifiers with probability λ, so one patient looks like two; MERGE relabels a random pair of participants to one pseudo-identifier, so two patients look like one. The last two mimic the observed Tier-1 failure modes and leave no field missing. These operators are controlled stress tests of identifiable provenance failure modes; they are not estimates of the prevalence or the distribution of such errors in clinical repositories, and we draw no conclusion about how often real cohorts sit at any given intensity. Only the grouping differs between the protocols compared; both use the same class-stratified assigner, for the reason given above. The experiment has two halves, and only the second needs a GPU. Contamination is a property of

the split, so the leakage every cell admits is measured without fitting anything; the AUROC each corrupted split then produces is measured by training one model per protocol per seed, 120 models in all.

Result the table in the main manuscript gives the matrix. The three mechanisms separate cleanly, and one of them is where the component layer stops being redundant. Under DROP the graph halves the damage at $\lambda = 0.1$ (16.8 straddling participants under subject-wise grouping against 8.4), falling to 31% prevented at total loss, which reproduces §S23 by a second route. Under SPLIT it prevents 50% at $\lambda = 0.1$ (a small absolute effect at that intensity, 1.2 straddling participants against 0.6) and 90% at $\lambda = 1$, where subject-wise grouping admits 61.6 straddling participants and the graph admits 6.4, with test-scan contamination falling from 64% to 11%. That gap is the framework's reason to exist: when one participant is dealt into two keys, `same_session`, `same_series_uid` and `same_acq_date` still join the halves, and subject-wise grouping has nothing left to join them with. Under MERGE neither protocol admits leakage at any intensity, because over-grouping two patients into one key is conservative for contamination: it destroys the participant count and the independence of the test partition without ever placing a patient on both sides. We report that null because it bounds the claim. The graph protects against identity *fragmentation*, not against every corruption, and a merged key damages a study in a way no splitter can repair.

The two leakage columns disagree in one cell, and the disagreement is instructive rather than inconsistent. At DROP $\lambda = 0.25$ the graph cuts straddling participants from 31.2 to 19.8 while the share of contaminated test scans rises from 0.364 to 0.395. The component layer builds fewer, larger groups, so a group that does straddle carries more scans across the boundary with it; counting participants and counting scans are different measurements of the same partition, and a protocol can improve one while losing the other. We print both columns rather than the one that reads better, and we compose the dose-response slope through the scan share, which is the conservative of the two (§S22).

What the prevented contamination is worth Preventing contamination is worth reporting only if the contamination was worth something, so we train one model per protocol per seed on each corrupted manifest and read the AUROC it produces (Table S12). The direction is consistent and the per-cell evidence is thin, and both halves of that sentence belong in the record.

Across the eight cells in which the two protocols emit different partitions, seven favour the component-safe grouping. The mean paired difference over those cells is +0.026 (SD 0.017); a one-sided sign test on the cell means gives $p = 0.035$ and a signed-rank test $W^+ = 35$, $p = 0.008$. Those two are summaries of one comparison and not independent confirmations of it, and both are descriptive rather than inferential: the cells share seeds and a base manifest, so they are not eight independent experiments. Only two cells, DROP $\lambda = 0.25$ and $\lambda = 0.5$, have an interval that excludes zero, and both exclude it by 0.001. Five seeds cannot resolve a per-cell effect of 0.02 to 0.04 AUROC when the paired SD runs from 0.024 to 0.079, so we claim a consistent direction and no per-cell effect size.

The magnitude is also smaller than the dose-response slope predicts. At SPLIT $\lambda = 1$ the graph removes 53 percentage points of test-scan contamination, from 64% to 11%, and Equation S1 would price that at roughly +0.066 AUROC against the +0.025 observed. The slope is fitted where contamination is substituted in and the training set thins as a result; corrupting provenance perturbs both sides of the split at once, and the two are not interchangeable. We report the discrepancy rather than the more favourable of the two numbers.

Two mechanisms deserve an explanation rather than a number. Under DROP $\lambda = 1$ the dif-

ference inverts to −0.004: with every participant identifier gone, both protocols admit near-total contamination, 99% of test scans under subject-wise grouping and 98% under the graph, and both saturate at 0.956 and 0.959. There is nothing left to protect, and the sign of a difference that small is noise, even though the table in the main manuscript still credits the graph with 31.3% of the straddling prevented in that cell: preventing leakage and being paid for it in AUROC are different claims, and this cell separates them. Under MERGE the difference is exactly 0.000 at every intensity because both protocols produce the same partition, so this is an identity and a determinism check, not an effect. The graph buys a modest amount of optimism back where provenance damage is the kind it can repair, and nothing at all where it is not.

S25 Clinical and field-level implications

At a fixed screening specificity of 0.90, ADNI sensitivity falls from 0.866 under random splitting to 0.535 under component-safe evaluation (§S21.0.1): 45.7 additional missed AD cases per 1,000 screened at the US 75–84 population prevalence anchor. A model whose sensitivity moves by a third between its benchmark and its honest evaluation is not a standalone screening tool. This gap arises from a design choice, the evaluation protocol, that is invisible in most published papers, which is the precise reason SPLITGUARD-AD reports the audit as a binary pre-training GO/NO-GO gate rather than as a post-hoc diagnostic. Two clinical-scope notes follow. *First*, CN-vs-AD with MCI excluded is the diagnostically easier cell of the AD imaging task. The cells that matter most for early-intervention decisions, CN-vs-MCI and stable-vs-converter MCI, are harder, and whether provenance-related inflation is larger there is an open empirical question: we did not run those tasks, and the direction is not obvious from what we did run. A smaller disease signal need not mean a larger measured gap if the leaky protocol is also less able to exploit identity. The frozen manifests released here support that experiment without rebuilding the cohort. *Second*, the headline absolute AUROC numbers (0.829 honest, 0.947 leaky) are floor estimates from a single 2D coronal-centre slice and should not be read as performance ceilings; the transferable quantity is the *gap*, not the level.

Beyond the headline number, two scoping points matter for clinical readers. First, structural MRI is not a standalone diagnostic tool for AD; in clinical practice it informs the dementia workup by excluding structural mimics, by demonstrating atrophy patterns consistent with neurodegeneration, and by supporting clinical and biomarker-based diagnosis within a research framework that defines the disease biologically rather than by imaging alone [37]. Nor is population screening for cognitive impairment itself an established benefit [38], which is a further reason the operating-point translation in §S21.0.1 is an arithmetic re-expression and not an argument for deployment, whatever the societal cost of dementia makes such a tool look worth [39]. A deep learning model evaluated on single slices labelled by a clinical rating cannot claim to diagnose Alzheimer's disease; it can only claim to classify images by that rating in a controlled benchmark setting, and SPLITGUARD-AD makes this distinction explicit through its clinical claim boundary and label-confidence fields. Second, lower leakage-safe performance should not be read as a failed model; it is a more honest estimate of what remains after removing patient-identity and acquisition shortcuts.

For redistributed public datasets, near-perfect internal scores are not interpretable unless the release documents provenance that makes a participant-safe split checkable: Tier 1 shows that a manifest built on a release's own identifier can pass every overlap check and still leave most of its test set contaminated. For metadata-rich cohorts such as OASIS-1, random slice splitting can still contaminate nearly every test subject, even when the resulting AUROC inflation is smaller and more seed-sensitive than on the redistributed benchmark. Future AD MRI studies should therefore

release split manifests, report subject overlap checks, document where both labels and identifiers come from, include calibration and sensitivity-at-fixed-specificity reporting, and validate on cohorts whose subjects, scanners, and preprocessing pipelines are independent of training.

S26 Limitations

- **The component layer is equivalent to subject grouping on intact cohorts, and validated beyond it only under synthetic provenance loss.** On the two source cohorts every connected component contains exactly one participant (Supplementary Table S5): the session, series-UID and acquisition-date rules fire, on 223, 209 and 222 ADNI pairs, but each joins a pair `same_subject` had already joined. The marginal over Protocol B is accordingly small and unstable: 0.008 [-0.033, +0.042] on the primary arm, and ranging from −0.026 to +0.026 across the six arms, with a sign inversion on the converter-inclusive one (the table in the main manuscript). No interval excludes zero in any of them, the one measured no-signal contrast of this kind is +0.094 (§S28), and we claim no AUROC benefit on intact cohorts. §S23 does establish a benefit, but on a regime we imposed rather than observed: it deletes identifiers uniformly at random from a cohort with complete provenance, whereas real redistributed benchmarks may lose them with structure our model does not reproduce: by site, by archive batch, or correlated with class. That measurement is also structural rather than predictive; it counts admitted leakage; its AUROC arm, trained on the same degraded manifests, gives 0.865 under subject-wise splitting against 0.829 under component-safe splitting at 10% loss. Tier 1 supplies the observed counterpart (§S20.0.1), and on it the component layer does not recover what the identifier lost, because the redistribution removed the session and series identifiers along with the participant one. A cohort whose damage leaves partial provenance for the graph to carry is still the most informative next experiment. The near-duplicate detector is now calibrated against recovered identity (precision 0.986 and adjacent-slice recall 0.246 at $d \leq 4$); it ships report-only because no threshold we measured is both safe and useful (§S15.0.2).

- **Every finding is a property of a 2D coronal-centre-slice representation.** The inflation gap, the dose-response slope, the provenance curve, the subgroup disparity and the clinical translation are all measured on one slice per volume, chosen to keep ADNI and OASIS-1 comparable; OASIS-1 additionally derives five slices per processed volume, so its operating-point metrics are small-sample and seed-sensitive. ResNet-18 carries the primary results, with a pre-specified DenseNet-121 arm replicating the headline gap. Whether these quantities transfer to volumetric models is now partly answered rather than open: the volumetric arm (128^3 volumes, 3D ResNet-18, `scripts/train_adni_3d.py`) runs against the same frozen partitions and puts the gap at +0.235 (§S19), so the identity shortcut survives whole-volume input and is in fact larger there. What does not transfer is the *level*: that network trains from scratch on 722 volumes while the 2D arms fine-tune ImageNet features, so its honest AUROC of 0.736 is a weak volumetric baseline and the arm settles the direction of the effect, not the achievable performance of 3D models. A pretrained or larger-cohort volumetric replication remains open. Multiple-instance aggregation over per-slice predictions and transformer backbones remain open, and reuse the same frozen manifests.

- **Label and identifier provenance differs by tier.** Tier-1 labels are CDR, but its participant identities are recovered by content matching rather than supplied, and we release the matching script against OASIS-1 rather than the mapping itself. Because all 200 Tier-1

participants are also in Tier 2, the two tiers are not independent replications of each other. On ADNI, 22% of labels (21% within the CN/AD subset, and 25.1% within the primary arm, which excludes the converter components the visit key covers best) are assigned through a 180-day date-proximity join rather than an exact visit key (§S16.0.3), and 28 magnetization-transfer T1 acquisitions (17 in the CN/AD universe) are non-MPRAGE; a pre-specified arm dropping them yields +0.128 [+0.097, +0.159] against +0.118 [+0.096, +0.145]. Restricting evaluation to the test scans whose label comes from an exact visit key gives a gap of +0.113; that tests the labels the gap is scored against, not the ones it was trained on, since the date-proximity scans remain in the training partition. The exact-linkage arm closes that half by retraining on the 841 exact-key scans alone (the table in the main manuscript), and the total gap is unchanged, so neither the labels the models learned from nor the labels they were scored against carry the result.

- **The intervals are descriptive, not inferential.** Five seeds support paired-seed bootstrap intervals that propagate between-seed variation only. The hierarchical subject × seed bootstrap, which also propagates participant-level sampling error and yields intervals 1.9–4.7× wider depending on the arm, is run on every arm. The three protocols also do not place comparable numbers of patients in the test partition (Supplementary Table S8): Protocol A averages 160.8 distinct participants per seed, Protocol B 48.0, and Protocol C 38.0, over test partitions of similar image count. Protocol C's AUROC is therefore estimated on roughly a quarter as many independent patients as Protocol A's, and part of its wider seed-to-seed spread reflects that rather than any property of the split rule. Subsampling participants does not correct for it, since AUROC is a rank statistic whose expectation barely moves with the number of clusters; the subject-level analysis above is the aggregation that does address it. The same asymmetry qualifies our recommendation of subject-level aggregation: averaging 5.35 predictions per patient under Protocol C removes substantially more variance than averaging 1.32 under Protocol A, so it does not neutralise the difference between protocols so much as redistribute it. Any single interval here should be read descriptively rather than as inference over a hypothetical larger seed population.

- **The secondary analyses support less than they appear to.** The identity probe uses one architecture and one probe family and reports a value that does not differ between leaky and honest protocols; it establishes that identity is decodable, across sessions as well as within them, not that it causes the gap. The sex-related disparity rests on five seeds and one cohort. The clinical translation is anchored to a single fixed-specificity operating point and to US-population prevalence, is computed on ADNI only, and is an arithmetic re-expression rather than a deployment estimate. The dose-response slope is specific to this cohort, this representation and the subject-substitution injection mechanism; the DenseNet-121 fit is noisier ($R^2 = 0.64$ against 0.85) and non-monotonic at $p = 1$, so we present it as a same-direction check rather than an independent confirmation, and we do not claim the slope generalises.

- **Disjointness is necessary but not sufficient, and our own gate demonstrates the gap.** The frozen ADNI splits pass every check the audit performs, zero subject, session and component overlap, while being stratified by component size (§S20.0.1). A split can be provably leakage-free and still be a poor split, and the gate as specified could not tell the difference. We regard this as the most useful negative result about the framework here, because the framework's own output produced it. The released audit now reports the component-size distribution per partition and warns when partitions are not drawn from comparable

participants; against the frozen manifests it fires immediately. It warns rather than blocks, deliberately: the manifests are frozen and every result rests on them. The size-balanced regeneration reported beside the headline shows the gap survives the correction. The same audit surfaces a second composition defect we did not fix: on the converter-inclusive arm the validation partition carries 20.7% AD images against 41.3% in training and 37.5% in test, in every seed, and the operating-point and cost-of-leakage thresholds are selected on that partition. The released audit now separates the two kinds of finding: partition disjointness is the only blocking check, and component-size and class-mix imbalance are reported as advisory warnings beside the verdict rather than folded into it. Promoting them to blocking is the first item of future work, and would retroactively fail the manifests this paper reports on, which is why it has not been done here.

The protocol also pays for its safety in validation size. Holding whole components together leaves the component-safe validation partition with about 22 participants, against 33 under subject-only splitting and 124 under a random split, which makes checkpoint selection noisy: with the disease signal permuted away, the best validation AUROC exceeds the test AUROC by +0.159 under component-safe splitting against +0.007 under subject-only. That is the no-signal worst case. On the runs this paper reports, where validation carries real signal, selection optimism is +0.026, +0.030 and −0.026 under the three protocols, every interval spans zero, and the spread is 0.055 – small, not ordered in the direction the objection requires, and negative for the component-safe arm. A fixed-epoch replication would settle it more directly and we did not run one. A practitioner adopting component-safe splitting on a cohort this size should nonetheless expect checkpoint selection to be noisy.

The gate certifies partition disjointness and nothing more. A pass does not imply site- or scanner-independent generalisation: the site–label association in Supplementary S7 survives the component-safe split, which is the central caveat of this work. Patient identity is one shortcut channel among several, and closing it leaves site, scanner, protocol, acquisition date and demographic structure open.

S27 Supporting tables and figures

The floats the main text does not reproduce, retained so that every quantity they report remains traceable.

S28 Patient reuse manufactures apparent discrimination

The gap of the previous section is real and robust, but on its own it cannot separate "the leaky protocol learned the disease better" from "the leaky protocol recognised its patients". Both produce a higher AUROC under random splitting, and no amount of replication across arms distinguishes them. We separate them by removing the disease.

Design The instrument is not new. Chaibub Neto *et al.* [40] introduced permutation tests for exactly this purpose, to separate a disease signature from a participant signature when records of one participant can reach both sides of a split, and label-shuffling controls of the same shape are now standard in neurophysiology benchmarks. We apply it here because the question it answers is the one the inflation gap leaves open, and because the protocols it compares are retrained rather than re-scored.

Diagnosis labels are permuted across participants and the three protocols are retrained at the frozen settings, five seeds, 15 epochs. The permutation is applied once over the whole cohort rather than inside each partition, so a participant who straddles a boundary carries the same permuted label on both sides: within-participant label consistency survives, the association between anatomy and diagnosis does not. That detail decides what the control can answer. Permuting inside each partition preserves per-partition class balance exactly and hands a straddling participant two different labels, which forecloses identity memorisation by construction and drives every protocol to chance for a reason that has nothing to do with the split.

Whether the permutation actually took effect is checked rather than assumed. Across the scored test rows the permuted label coincides with the participant's true diagnosis 48.3%, 55.1% and 50.3% of the time under the three protocols. That is the chance rate, and it is the precondition for reading anything below as patient recall: had the true labels survived anywhere in the pipeline, the random arm would still score high and the result would read as a stronger version of the same finding rather than as a broken one.

Result With no disease signal left to learn, the random split still reaches an AUROC of 0.878 $[0.837, 0.919]$. The two honest protocols sit at chance: 0.521 $[0.427, 0.614]$ under subject-only splitting and 0.427 $[0.320, 0.533]$ under component-safe splitting, both intervals covering 0.5. The seed-paired contrast is $+0.451$ $[+0.357, +0.546]$, positive on all five seeds.

The training curves say where it comes from. All three protocols drive the permuted training loss to a comparably low value, 0.068, 0.037 and 0.045 at the final epoch, so all three memorise the training set about equally well. Only the random split's memorisation transfers: its final validation AUROC is 0.863 against 0.453 and 0.481 for the honest protocols, on a label that carries no information about any image. This is the shortcut being taken, measured rather than inferred, and it is the evidence the probe could not supply.

What the number is not The permuted gap of $+0.451$ is not the inflation under real labels, which is $+0.118$. It measures the capacity of the memorisation channel when nothing else is learnable. Under real labels the honest protocols score around 0.83 because there is a real signal to learn, and the leaky protocol's score is bounded above; the channel measured here is one contributor to that score rather than the whole of it. We report it as evidence that the channel exists and is large, not as a revised effect size.

A measured noise floor The control also prices the paper's null result. Under permutation the subject-only and component-safe arms differ by $+0.094$ $[+0.013, +0.175]$, an interval excluding zero, between two protocols that both have nothing to learn and are therefore both at chance. It cannot be a difference in what was learned, and its sign has a mechanism worth naming, because it is not specific to permuted labels. The trainer keeps the epoch with the best validation AUROC. With the disease signal removed, the honest arms' validation partitions carry none either, so that maximum is selection on noise, and the smaller the partition the further it overshoots: validation optimism, the best validation AUROC minus the test AUROC, is $+0.007$ under subject-only splitting on a validation partition averaging 33 participants, and $+0.159$ under component-safe splitting on one averaging 22. The leaky arm shows -0.008 on a validation partition averaging 124 participants, because it holds the same patients as training: the signal it selects on is real memorisation, so there is nothing to overshoot.

Two things follow. A contrast of $+0.094$ arises here between protocols with nothing to learn, an order of magnitude above the 0.008 component marginal the main experiment reports (the table

in the main manuscript), which is why we decline to read that marginal as an effect and decline to settle it with more seeds of the same kind. And a component-safe design pays for its safety in validation size: 22 participants is too few to select a checkpoint on, with permuted labels or real ones, which is a cost of the protocol we had not previously quantified and which any user of it inherits. It is the scale of apparent protocol-to-protocol difference this design admits at five seeds, and it is why we decline to read the component marginal as an effect and decline to settle it by adding seeds of the same kind.

Table S11: Leakage and shortcut taxonomy for Alzheimer's MRI deep learning. Risk levels are indicative for a redistributed public 2D benchmark and reflect the typical exposure of a CN-vs-AD slice-level classifier; absolute risk in any new study depends on cohort-specific structure.

Mode	**Risk**	**Description**
Subject leakage	Critical	Same patient in train and test.
Session leakage	High	Same visit/acquisition across splits.
Longitudinal leakage	High	Same patient at different time points.
Slice leakage	High	2D slices from one volume cross splits.
Near-duplicate	Moderate	Perceptually similar derived images.
Source leakage	Moderate	Dataset/collection provenance artifact.
Site/scanner	Moderate	Scanner, field strength, protocol artifact.
Demographic shortcut	Low–Mod	Age/sex imbalance correlated with label.
Clinical-label shortcut	Low	Advanced atrophy makes task trivially easy.

S29 Result tables generated from the analysis artefacts

These tables are written by the analysis scripts rather than by hand, and are reproduced here in full. The main text quotes the quantities it needs from them.

Table S14 gives the ADNI1 cohort composition.
Table S15 gives the primary-arm protocol comparison.
Table S16 gives every robustness arm.
Table S17 gives subject-level aggregation.
Table S18 gives the OASIS-1 replication.
Table S19 gives the Tier-1 recovered-identity audit.
Table S20 gives the screening operating-point sensitivity analysis.

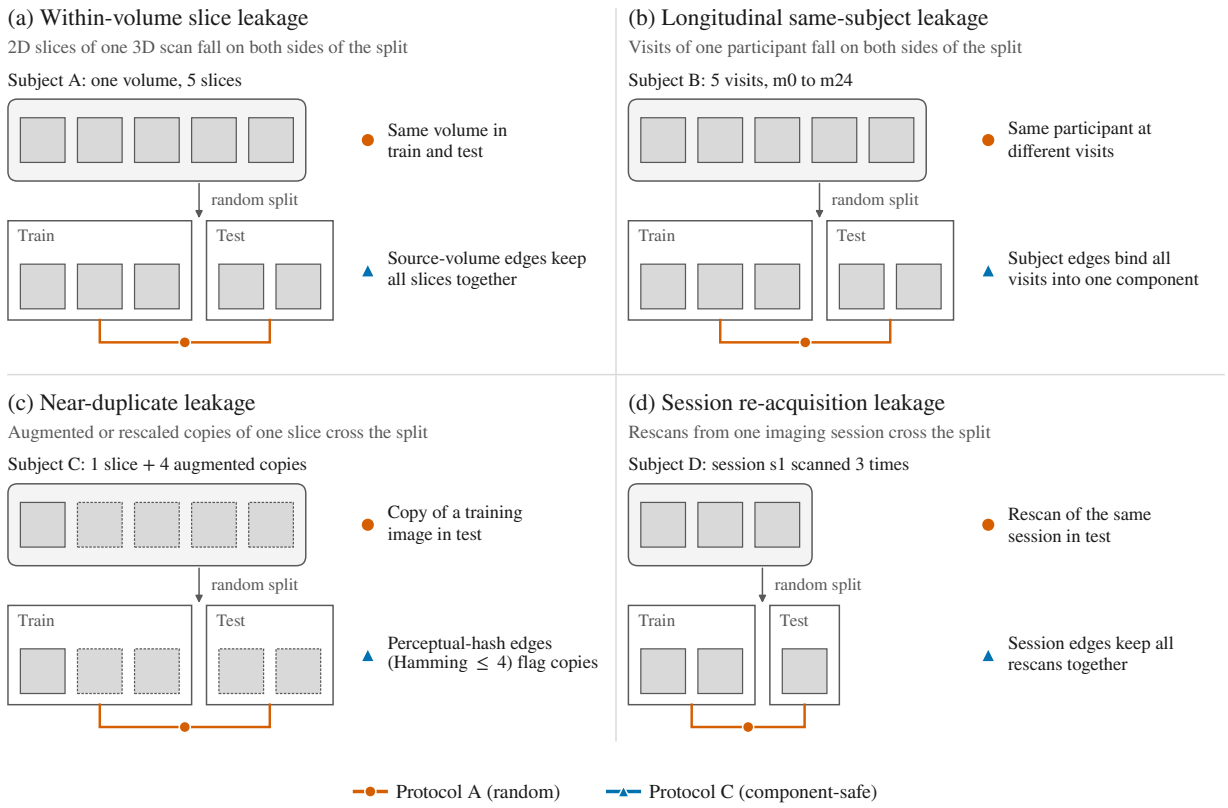


Figure S2: Four leakage modes in slice-based AD MRI benchmarks, and the SplitGuard-AD edge type that addresses each: (a) source-volume (series-UID) edges, (b) subject-identifier edges, (c) near-duplicate dHash (Hamming ≤ 4, reported not merged under the default policy), (d) session-identifier edges. In each panel the bracket and circle mark the leak a random split (Protocol A) produces, and the triangle the edge that prevents it under component-safe splitting (Protocol C). Connected components under the union of the blocking edges are the indivisible split units. **This figure describes the framework's design space, not its empirical exercise in this paper.** On the two source cohorts every connected component contains exactly one participant; on Tier 1 the components built from the release's own identifier contain up to three participants each, because that identifier is not a participant identifier (§S16.0.1). Modes (a) and (d) do fire on ADNI (209 source-volume and 223 session pairs, contributing to 115 of the 382 components), but each joined a pair that mode (b) had already joined, so no component differs from a plain subject grouping; mode (c) is detected and reported but not merged under the default policy. All four are implemented and auditable; none is load-bearing on these cohorts, which is why the component layer's measured marginal over subject-wise splitting is small (§S20.0.1, Supplementary Table S5). Demonstrating them requires a cohort with re-acquisition or augmentation structure that subject identity does not already capture; we name this as the primary validation gap in §S26.

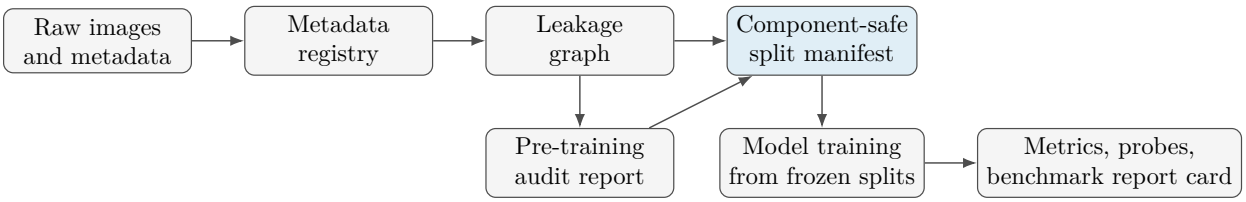


Figure S3: The SplitGuard-AD pipeline: from raw images to audit-verified split manifests and benchmark report cards. Modules are executed sequentially; each produces a versioned artefact consumed by the next.

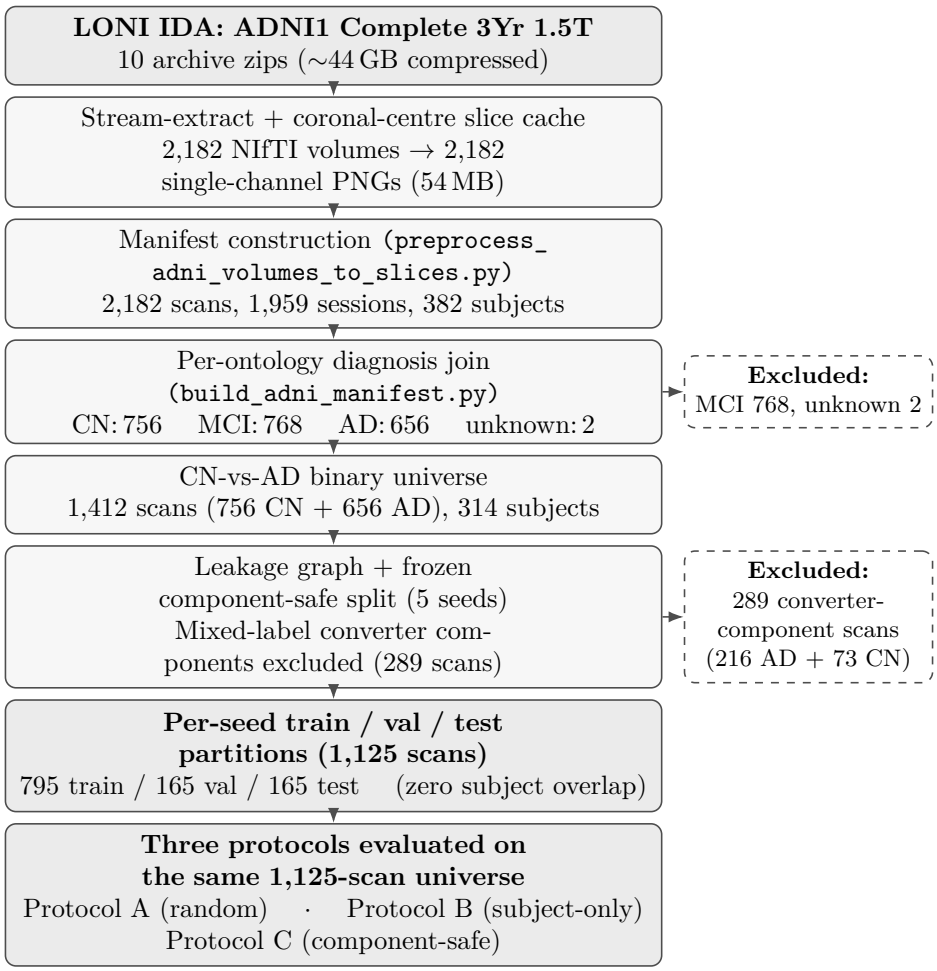
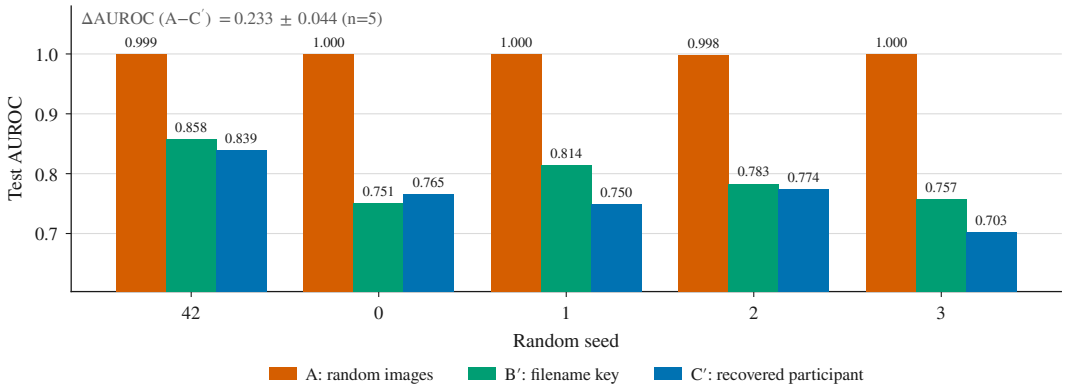
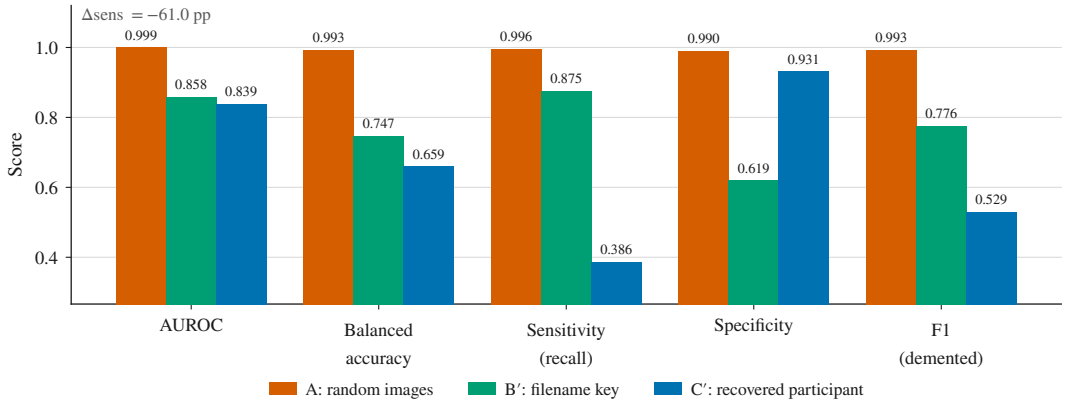


Figure S4: ADNI1: Complete 3Yr 1.5T data flow from LONI IDA archive download to per-seed train/validation/test partitions. Each transition annotates the row count and the script responsible (`preprocess_adni_volumes_to_slices.py`, `build_adni_manifest.py`, `make_adni_splitguard_split.py`). Dashed boxes on the right record exclusions: MCI / unknown diagnoses excluded from the CN-vs-AD task by construction, and 289 CN/AD scans (216 AD + 73 CN) from mixed-label longitudinal converter components excluded from the primary arm and re-included in the sensitivity arm (§S19).



(a) Per-seed test AUROC under the three grouping rules.



(b) All test-set metrics at seed = 42.

Figure S5: Tier-1 under random splitting (A, vermilion), the release's filename key (B′, green) and recovered participant identity (C′, blue): (a) seed stability, and (b) the per-metric breakdown. Panels are stacked vertically rather than side-by-side so that the in-figure tick labels and bar annotations remain legible at print size.

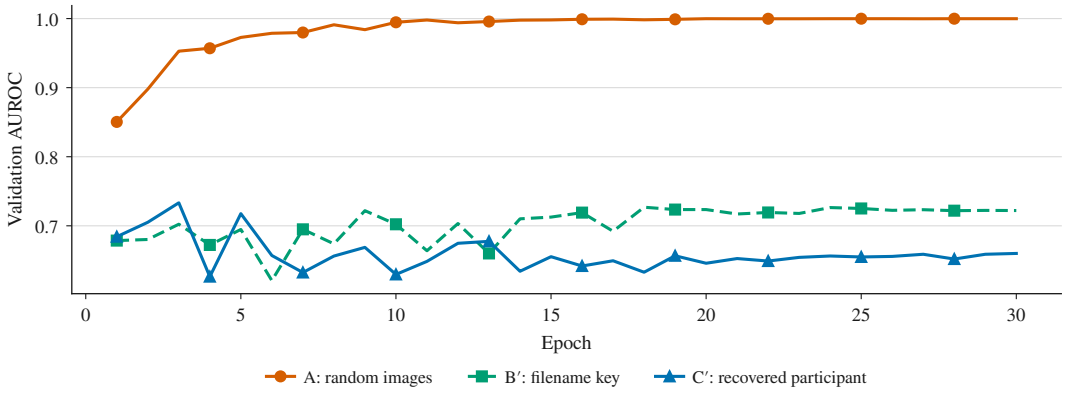


Figure S6: Validation AUROC per epoch at seed = 42 for Protocol A (random, vermilion), Protocol B′ (filename key, green) and Protocol C′ (recovered participant, blue). The vertical distance between the first and last curves is the inflation the release's own identifier cannot remove.

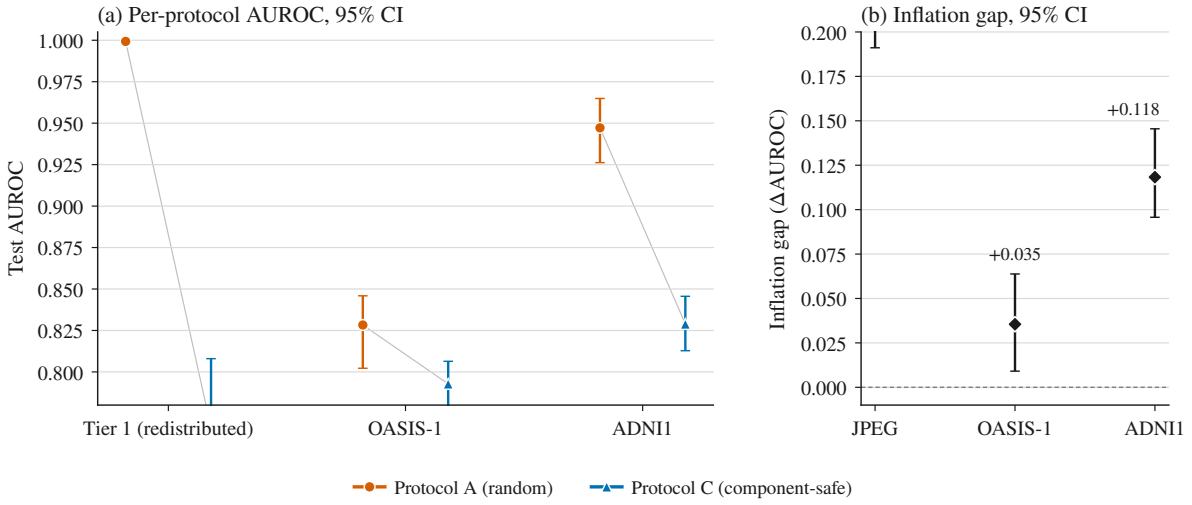


Figure S7: Direction-preserved inflation gap on the two source cohorts and on the redistributed benchmark (ResNet-18). (a) Per-protocol test AUROC (random versus participant-safe) for each cohort; error bars are paired-seed bootstrap intervals (B = 10,000, n = 5 seeds per cohort; descriptive of seed-level variability, see "Statistical conventions" in the main manuscript). Grey lines connect the two protocols within each cohort. (b) The inflation gap ΔAUROC with the same descriptive intervals. On Tier 1 the participant-safe protocol is the identity recovered in §S16.0.1, not the release's own identifier, which is why its gap is larger here than the one a reader of that benchmark could measure. DenseNet-121 replications are reported in §S18 (OASIS-1) and §S19 (ADNI); Tier 1 was run on ResNet-18 only; the figure shows ResNet-18 for visual parity with the rest of the paper.

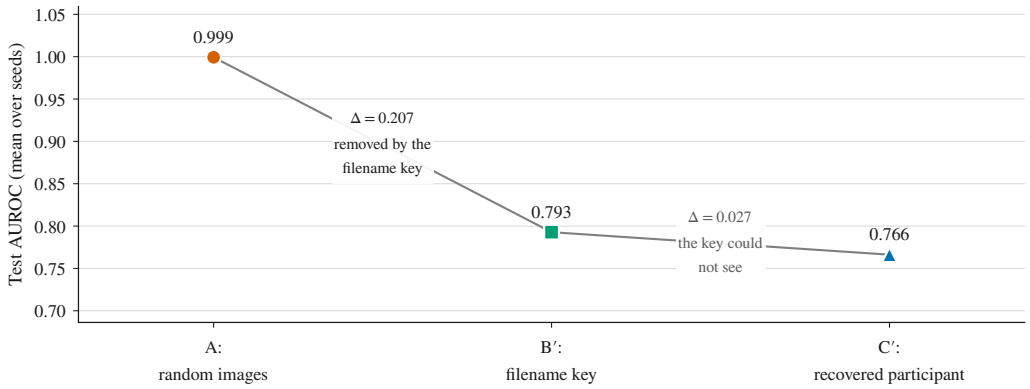


Figure S8: Three-point degradation on Tier 1. Each step changes only what the split groups by: random images, then the filename key the release ships, then the participant identity recovered from OASIS-1. The first step is what an audit run inside the release can achieve; the second is what its lost provenance hides.

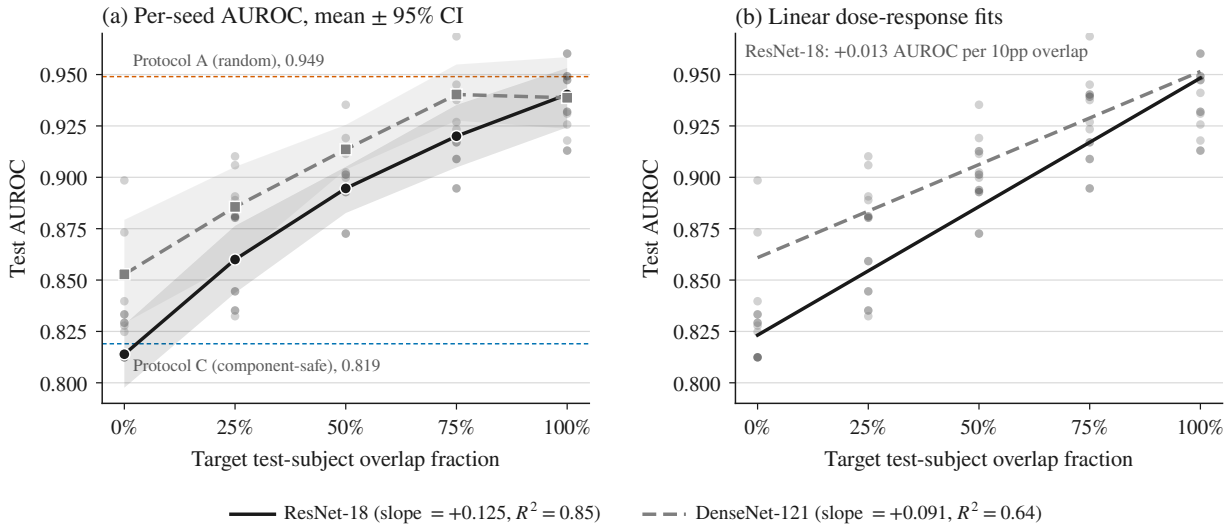


Figure S9: ADNI leakage dose-response on the converter-inclusive sensitivity arm. (a) Per-seed test AUROCs (small dots), seed-averaged mean (markers + line) and 95% paired-bootstrap CI band for each architecture as a function of the target test-subject overlap fraction p; ResNet-18 is drawn in black and DenseNet-121 in grey (dashed), and the reference lines take the colour of the protocol they mark. Reference lines mark the *primary-arm* Protocol C component-safe baseline (0.829) and Protocol A leaky benchmark (0.947) from §S19, shown for cross-arm comparability. The matrix itself is built on the converter-inclusive arm, whose own levels are 0.815 and 0.957; the fitted endpoints are 0.823 and 0.948 (panel a). (b) OLS linear fits to the per-seed points show the dose-response slope; the ResNet-18 slope of +0.125 AUROC per unit overlap is a cohort- and architecture-specific calibration point on this ADNI universe and this 2D-slice representation, not a claim that the slope generalises to other cohorts or 3D representations.

Table S12: AUROC consequence of each corruption cell on the ADNI1 primary arm. Each row trains one model per protocol per seed on the *same* corrupted manifest, so the comparison is paired: the difference column and its interval are over the five per-seed differences (paired t, four degrees of freedom), not over the two marginal means. A positive difference means the component-safe grouping scored *lower*, that is, it declined optimism the subject-wise grouping accepted. The last column counts seeds in which it did. MERGE is an identity rather than a null result: when two participants share one key both protocols emit the same partition, so the same model is trained twice, and the exact zeros double as a determinism check on the pipeline.

Corruption	λ	**Prot. B**	**Prot. C**	**Paired Δ**	**95% CI**	**Seeds**
DROP	0.1	0.847	0.814	+0.034	[−0.062, +0.129]	4/5
	0.25	0.857	0.812	+0.045	[+0.001, +0.089]	4/5
	0.5	0.945	0.902	+0.043	[+0.001, +0.084]	5/5
	1	0.956	0.959	−0.004	[−0.034, +0.027]	3/5
SPLIT	0.1	0.850	0.836	+0.014	[−0.035, +0.064]	3/5
	0.25	0.868	0.853	+0.015	[−0.024, +0.053]	4/5
	0.5	0.845	0.806	+0.038	[−0.027, +0.104]	4/5
	1	0.877	0.852	+0.025	[−0.073, +0.124]	3/5
MERGE	0.1	0.841	0.841	+0.000	[+0.000, +0.000]	0/5
	0.25	0.806	0.806	+0.000	[+0.000, +0.000]	0/5
	0.5	0.815	0.815	+0.000	[+0.000, +0.000]	0/5
	1	0.808	0.808	+0.000	[+0.000, +0.000]	0/5

Table S13: SplitGuard-AD benchmark report card for the ADNI1 primary-arm ResNet-18 model under the component-safe split (Protocol C), seed means over five seeds. The card illustrates the reporting format the framework produces; it is not a clinical-deployment artefact. PPV/NPV rows use the same prevalence anchors as §S21.0.1.

SplitGuard-AD Benchmark Report Card	
Model Claim Boundary	CN versus AD from one 2D coronal-centre slice per scan, labels from the ADNI diagnosis table. Not intended for standalone clinical diagnosis.
Identifier provenance	ADNI `PTID`, session and series identifiers, all present for every scan.
Data Safety Audit *(blocking checks)*	
Participant Overlap	0% (✓Pass)
Session Overlap	0% (✓Pass)
Connected Component Overlap	0% (✓Pass)
Reported, non-blocking	
Label join	25.1% of this arm's labels by 180-day date proximity; the gap is unchanged on the exact-key subset (§S16.0.3)
Site–label association	Cramér's $V = 0.54$ in test (Supplementary Table S7)
Component-size balance	Warning: partitions differ in scans per participant (§S20.0.1)
Debiased Performance Estimates (Held-out Test, $n \approx$ 165 per seed)	
Sensitivity	65.5%
Specificity	81.0%
AUROC	0.829
Balanced Accuracy	73.2%
Prevalence-conditioned predictive values (illustrative)	
PPV @ 13.8% (population) [1]	35.6%
NPV @ 13.8% (population)	93.6%
PPV @ 58.9% (memory clinic) [2]	83.2%
NPV @ 58.9% (memory clinic)	62.1%

Computed from sensitivity/specificity above; predictive values are prevalence-dependent and tracked at the same two anchors as §S21.0.1.

Table S14: ADNI1: Complete 3Yr 1.5T cohort demographics. Subject-level summary; subjects are assigned to a diagnosis group by highest-priority label observed across all visits (AD > CN > MCI). Age computed at each subject's earliest scan from `PTDOBYY`; baseline MMSE is each subject's earliest non-missing `MMSCORE`.

	CN	**MCI**	**AD**	**Total**
Subjects (n)	142	68	172	382
Scans (n)	794	442	946	2182
Age, years (mean ± SD)	75.7 ± 5.4	75.1 ± 7.1	75.0 ± 7.3	n/a
Female (n, %)	68 (48%)	15 (22%)	76 (44%)	n/a
Baseline MMSE (mean ± SD)	29.2 ± 1.0	27.5 ± 1.7	24.7 ± 2.5	n/a

Table S15: ADNI inflation gap, 5 seeds, 15 epochs, ResNet-18, coronal centre slice. Numbers are mean AUROC with paired-seed bootstrap 95% CI (B=10,000). "Test contam." = mean % of test subjects also appearing in the train partition. The lower panel drops the 28 magnetization-transfer T1 scans for a sequence-homogeneity sensitivity check.

Protocol	AUROC [95% CI]	n_{test}	Test contam.
All scans (MT1 included):			
Random (leaky)	0.947 [0.926, 0.965]	169	99.7%
Subject-only	0.837 [0.810, 0.862]	169	0.0%
Component-safe (SplitGuard)	0.829 [0.813, 0.846]	165	0.0%
Total gap (random − component-safe)	0.118 [0.096, 0.145]		
Subject-leakage (random − subject-only)	0.110 [0.086, 0.139]		
Component-marginal (subject-only − comp.-safe)	0.008 [-0.033, 0.042]		
MT1 acquisitions excluded:			
Random (leaky)	0.960 [0.947, 0.973]	168	99.7%
Subject-only	0.833 [0.805, 0.857]	168	0.0%
Component-safe (SplitGuard)	0.831 [0.810, 0.855]	163	0.0%
Total gap (random − component-safe)	0.128 [0.097, 0.159]		
Subject-leakage (random − subject-only)	0.127 [0.103, 0.154]		
Component-marginal (subject-only − comp.-safe)	0.002 [-0.017, 0.028]		

Table S16: ADNI robustness arms; the primary arm excludes converters. Columns are the total gap (Protocol A−C), which is the paper's headline, the subject part (A−B) and the component marginal (B−C), which is its null result; all three are measured on the same runs. Intervals are paired-seed bootstrap at $B = 10{,}000$.

Arm	**A−C**	**A−B**	**B−C**
Primary	$+0.118$ $[+0.096, +0.145]$	$+0.110$ $[+0.086, +0.139]$	$+0.008$ $[-0.033, +0.042]$
Converter-inclusive	$+0.142$ $[+0.131, +0.158]$	$+0.168$ $[+0.105, +0.232]$	-0.026 $[-0.100, +0.048]$
MT1 excluded	$+0.128$ $[+0.097, +0.159]$	$+0.127$ $[+0.103, +0.154]$	$+0.002$ $[-0.017, +0.028]$
DenseNet-121	$+0.146$ $[+0.134, +0.161]$	$+0.136$ $[+0.107, +0.165]$	$+0.011$ $[-0.026, +0.053]$
Size-balanced	$+0.136$ $[+0.098, +0.159]$	$+0.110$ $[+0.086, +0.139]$	$+0.026$ $[-0.040, +0.068]$
Exact label linkage	$+0.117$ $[+0.072, +0.166]$	$+0.093$ $[+0.039, +0.136]$	$+0.024$ $[-0.017, +0.065]$

Table S17: ADNI image-level against subject-level AUROC on the converter-inclusive arm (five seeds). Both columns are AUROC with a paired-seed bootstrap interval at $B = 10{,}000$. Subject-level numbers average each participant's per-image predictions before computing AUROC.

Protocol	**Image-level**	**Subject-level**
Random (leaky)	0.957 [0.946, 0.969]	0.966 [0.962, 0.969]
Subject-only	0.789 [0.730, 0.851]	0.818 [0.771, 0.873]
Component-safe (SplitGuard-AD)	0.815 [0.795, 0.835]	0.829 [0.794, 0.858]
Inflation gap (Protocol A − Protocol C):		
Image-level	+0.142 [+0.131, +0.158], positive in 5/5 seeds	
Subject-level	+0.137 [+0.108, +0.173], positive in 5/5 seeds	

Table S18: OASIS-1 replication: random slice split against the participant-safe SplitGuard-AD split, mean ± SD over five ResNet-18 seeds at 20 epochs.

Metric	**Protocol A**	**Protocol C**	**Gap (A−C)**
AUROC	0.828 ± 0.029	0.793 ± 0.020	0.035 ± 0.037
Balanced accuracy	0.712 ± 0.027	0.704 ± 0.024	0.008 ± 0.043
F1 (demented)	0.650 ± 0.069	0.683 ± 0.015	-0.033 ± 0.076
Sensitivity	0.687 ± 0.225	0.771 ± 0.137	-0.084 ± 0.220
Specificity	0.738 ± 0.174	0.638 ± 0.179	0.100 ± 0.185
Test subjects leaking	99.54%	0%	n/a

Table S19: Tier-1 test AUROC under the three grouping rules, five seeds. Protocol B′ groups by the filename key the release ships, which is the best an audit inside the release can do; Protocol C′ groups by the participant identity recovered from OASIS-1. The gap column is A−C′.

Seed	**A: random**	**B′: filename key**	**C′: participant**	**Gap (A−C′)**
42	0.9994	0.8581	0.8387	+0.161
0	0.9999	0.7508	0.7655	+0.234
1	0.9996	0.8142	0.7496	+0.250
2	0.9976	0.7834	0.7743	+0.223
3	0.9999	0.7574	0.7030	+0.297
Mean ± SD	0.999 ± 0.001	0.793 ± 0.044	0.766 ± 0.049	+0.233

Table S20: Operating-point sensitivity table for the ADNI converter-inclusive arm. **Every threshold is selected on the validation partition and applied to test**; no operating point is chosen by scanning the test ROC. The fixed-rate rows are mean ± SD across five seeds and the Youden rows are seed means, with Δ the Protocol A minus Protocol C difference.

Operating point	Random	Subject-only	Component-safe	Δ
Sensitivity at fixed specificity				
Spec = 0.80	0.919 ± 0.028	0.717 ± 0.122	0.649 ± 0.084	+0.270
Spec = 0.85	0.901 ± 0.024	0.640 ± 0.109	0.616 ± 0.100	+0.285
Spec = 0.90	0.866 ± 0.046	0.551 ± 0.105	0.535 ± 0.103	+0.331
Spec = 0.95	0.810 ± 0.057	0.382 ± 0.055	0.386 ± 0.099	+0.424
Specificity at fixed sensitivity				
Sens = 0.85	0.948 ± 0.028	0.567 ± 0.114	0.563 ± 0.108	+0.385
Sens = 0.90	0.898 ± 0.061	0.439 ± 0.124	0.488 ± 0.141	+0.410
Sens = 0.95	0.782 ± 0.125	0.299 ± 0.111	0.403 ± 0.162	+0.379
Youden-optimal threshold (selected on validation)				
J	0.785	0.458	0.533	+0.252
Sens at J	0.836	0.768	0.682	+0.153
Spec at J	0.950	0.690	0.851	+0.099